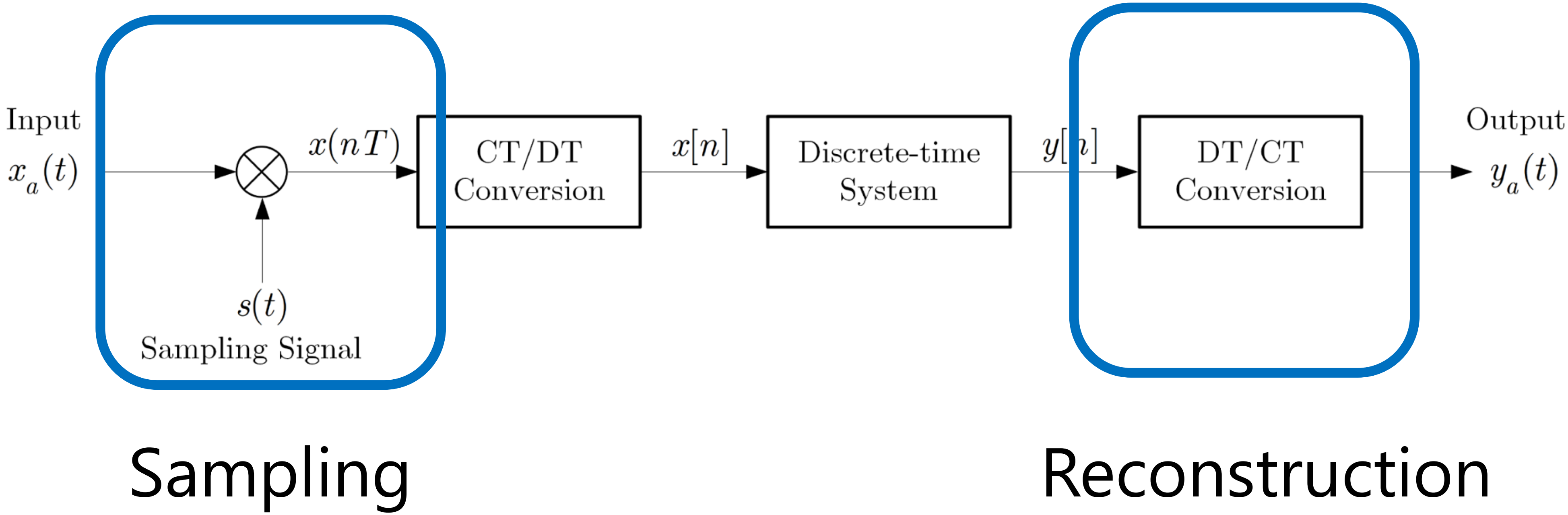
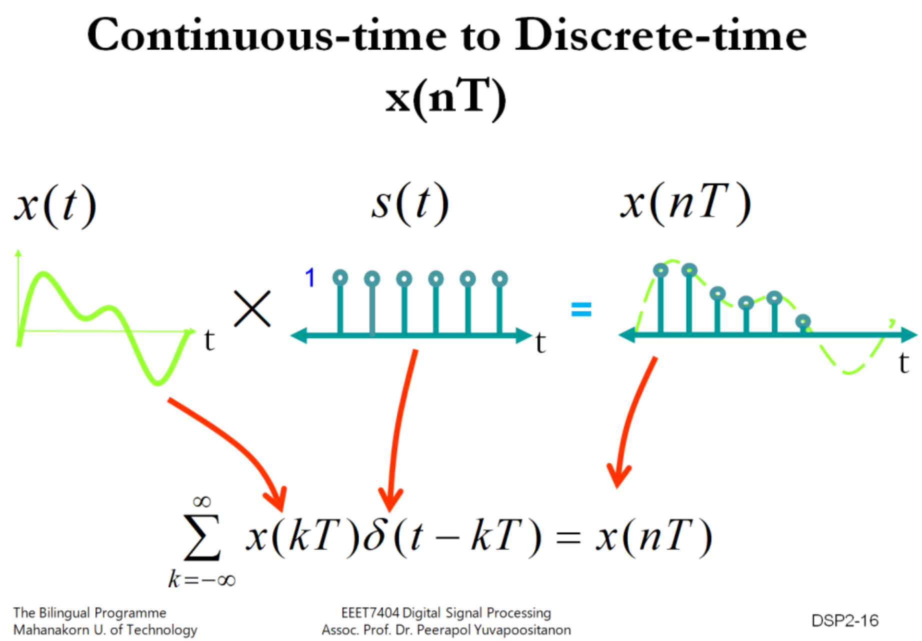
3
The Discrete-Time Fourier Analysis and Transform

DSP Building Blocks



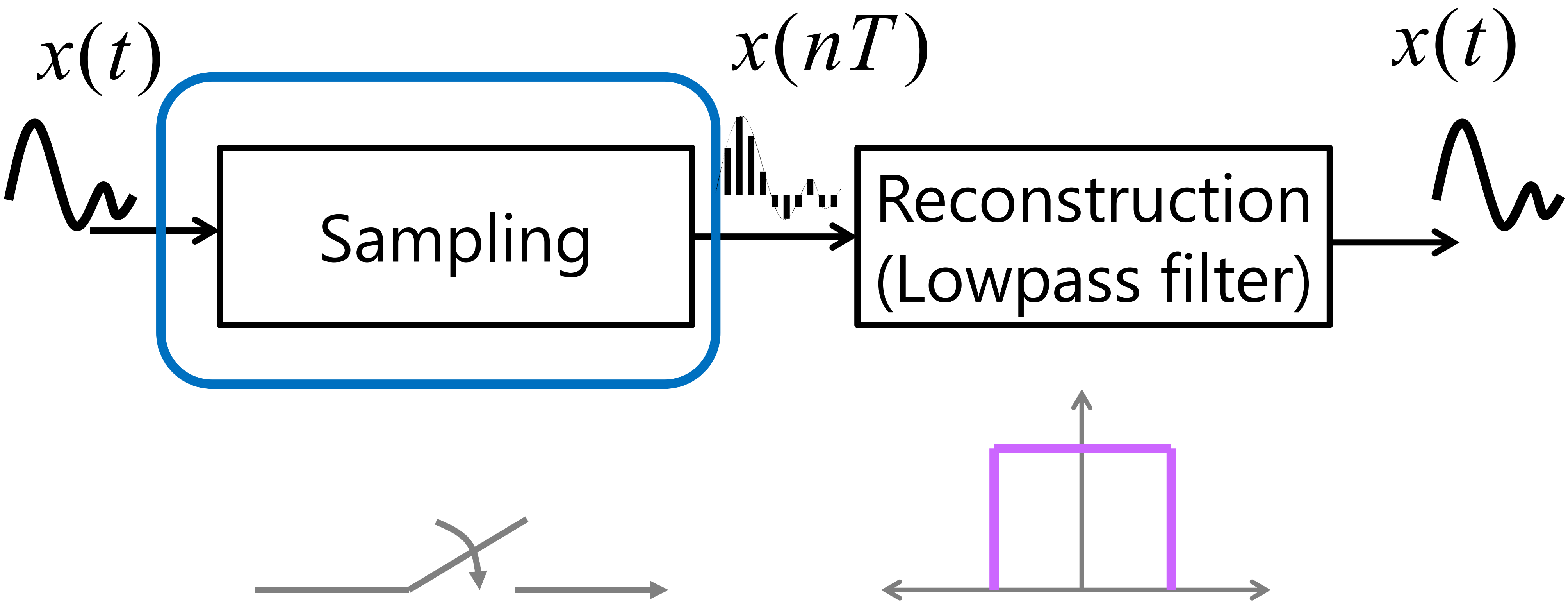
- From Ch.2



$$x(nT) = x(t)\sum_{k=-\infty}^{\infty}\delta(t-kT)$$

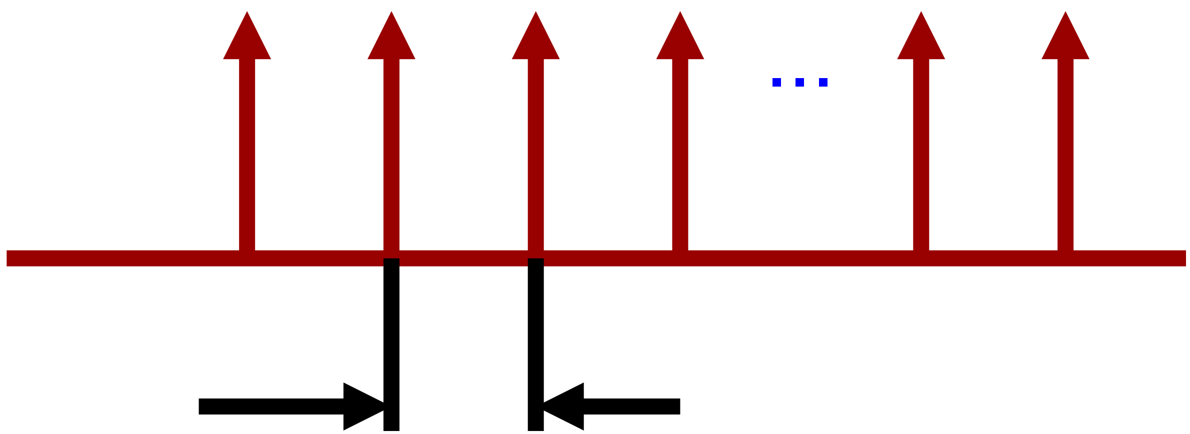
Sampling and Reconstruction

- Convert analogue signals to discrete-time format.



Sampling Signal

- Let us consider a (analogue) sampling signal with sampling frequency fs



$$T = \frac{1}{f_s}$$

Fs = f

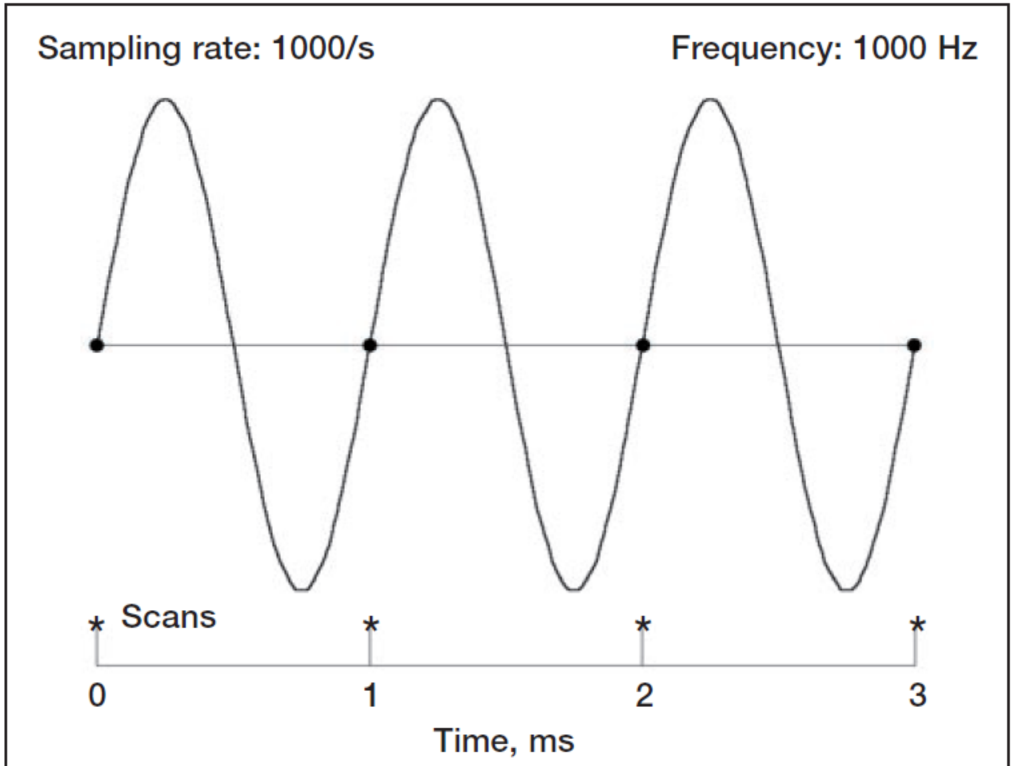


Figure 1

Fs= 1.33f

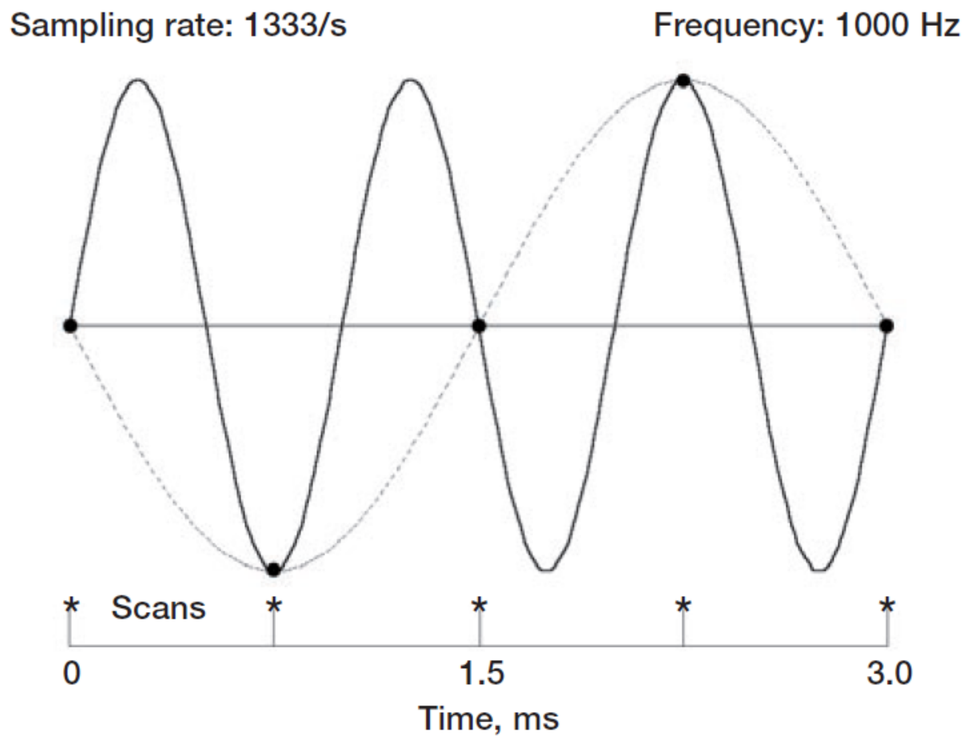


Figure 2

Fs=2f

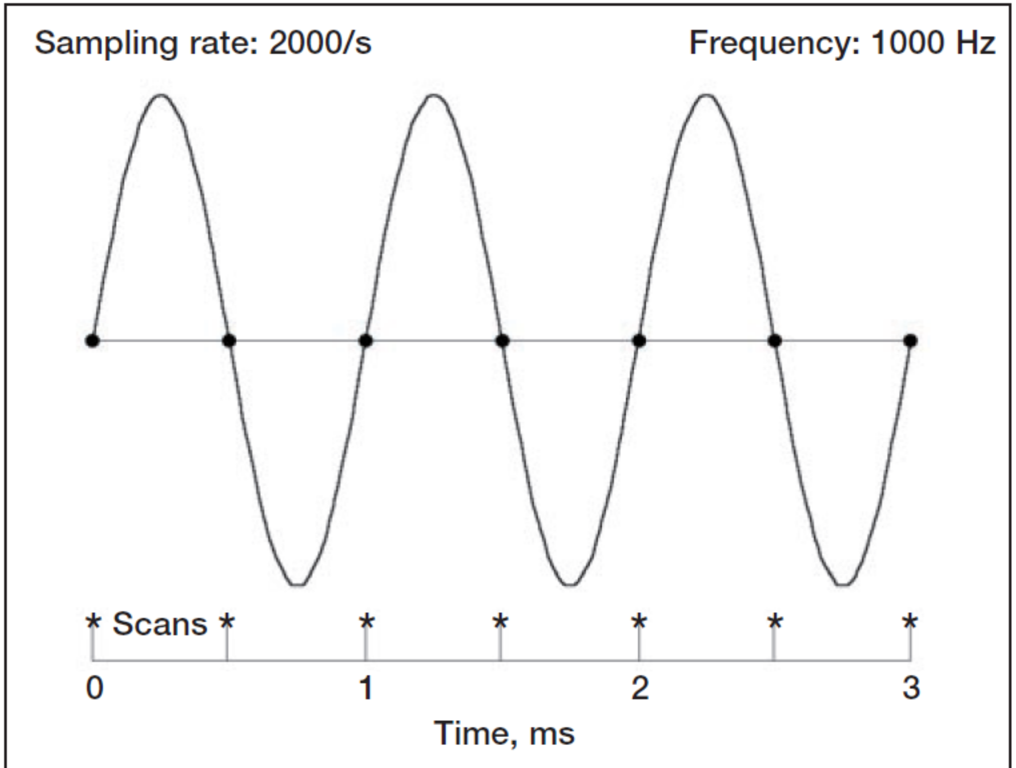


Figure 3

Fs=4f

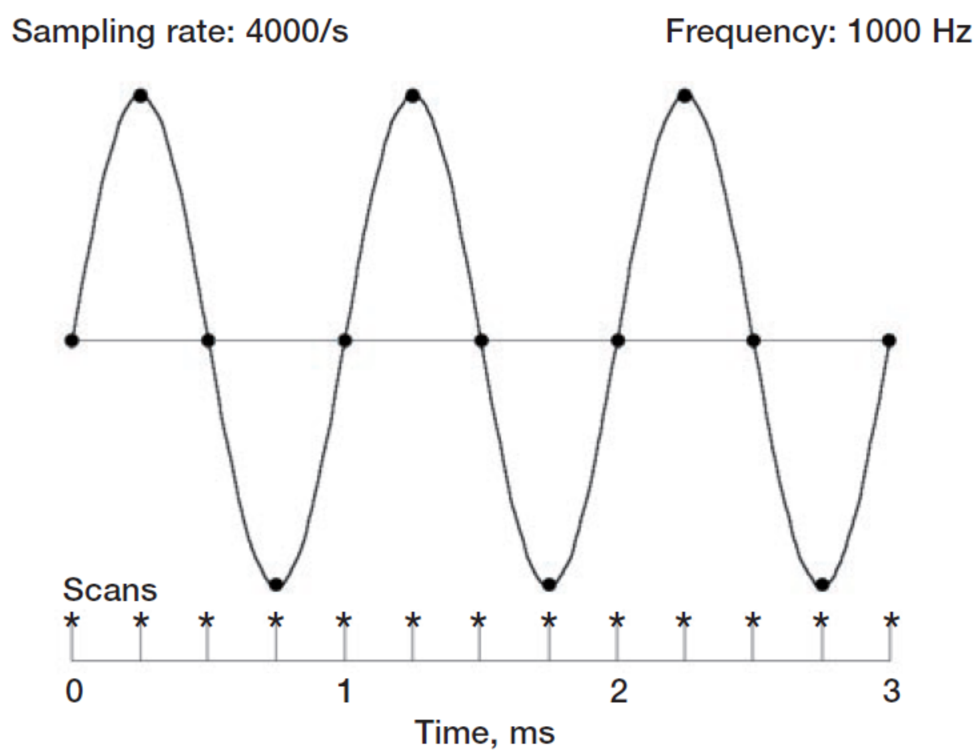


Figure 4

Example 3.19: Sampling Test

- Suppose we are sampling a sinusoidal signal x(t) with the fs =1000 Hz sampling signal where

$$x(t) = \cos(500\pi t)$$

$$= \cos(2\pi(250)t)$$

$$f_a = 250 \text{ Hz}$$

which means fa is **less than** the sampling signal fs.

(1) $x(n) = \cos(0.5\pi n)$

- From $f_a = 250\ \text{Hz}$

$$\omega = \Omega T = \Omega \frac{1}{f_s}$$

$$\omega = 2\pi f_a \frac{1}{f_s} = 500\pi \frac{1}{1000} = 0.5\pi$$

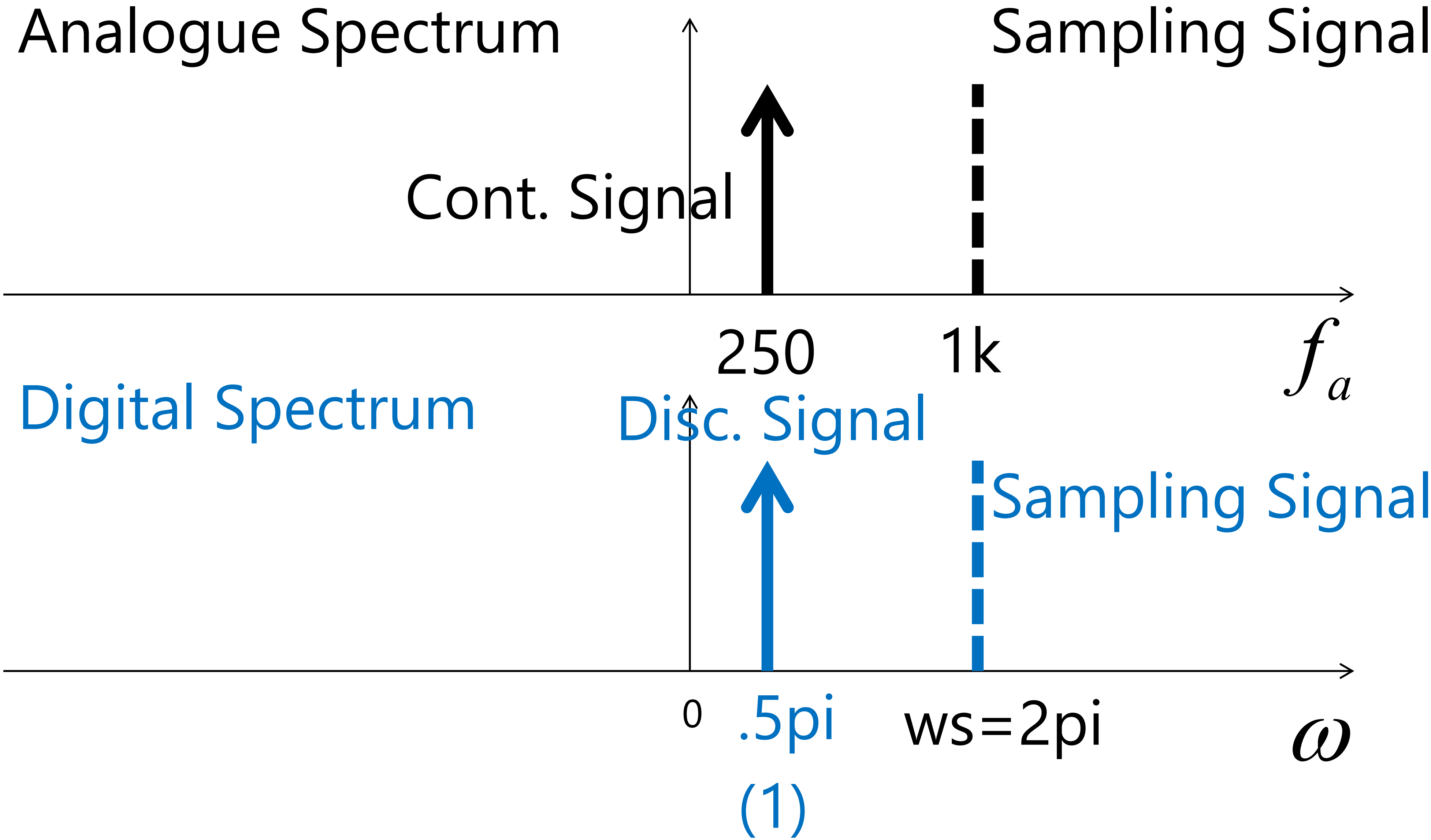
- So we have

$$x(n) = \cos(0.5\pi n)$$

The Sampling frequency in Digital domain is 2pi

- For the digital frequency domain, the sampling frequency is always

$$\begin{aligned}\omega_s &= \Omega T \\ &= 2\pi f_s \frac{1}{f_s} \\ &= 2\pi \quad \text{radians/sample}\end{aligned}$$


Analogue Spectrum
Sampling Signal
Cont. Signal
250
1k
f_a
Digital Spectrum
Disc. Signal
Sampling Signal
0
.5pi
(1)
ws=2pi
ω

Frequencies Replicas (Copies)

- However, when sampling x(t) in the analogue domain, **we get more than one digital frequencies in the digital spectrum!**
- Let us find three frequency replicas apart from cos(.5*pi*n) produced from sampling a 250Hz x(t) with fs=1000Hz.

(2) $x_2(n) = \cos(-1.5\pi n)$

- We can show that

$$\cos(-1.5\pi n) = \cos((-2+.5)\pi n)$$

$$= \cos(-2\pi n + .5\pi n)$$

$$= \cos(-2\pi n)\cos(.5\pi n)$$

$$- \sin(-2\pi n)\sin(.5\pi n)$$

$$\therefore \cos(-1.5\pi n) = \cos(.5\pi n)$$

(3) $x_3(n) = \cos(-3.5\pi n)$

- Next, we can show that

$$
\begin{aligned}
\cos(-3.5\pi n) &= \cos((-4+.5)\pi n) \\
&= \cos(-4\pi n + .5\pi n) \\
&= \cos(-4\pi n)\cos(.5\pi n) \\
&\quad - \sin(-4\pi n)\sin(.5\pi n)
\end{aligned}
$$

$$\therefore \cos(-3.5\pi n) = \cos(.5\pi n)$$

$$(4) \quad x_4(n) = \cos(2.5\pi n)$$

- Finally, we can show that

$$\begin{aligned}\cos(2.5\pi n) &= \cos((2+.5)\pi n)\\ &= \cos(2\pi n + .5\pi n)\\ &= \cos(2\pi n)\cos(.5\pi n)\\ &\quad - \sin(2\pi n)\sin(.5\pi n)\end{aligned}$$

$$\therefore \cos(2.5\pi n) = \cos(.5\pi n)$$

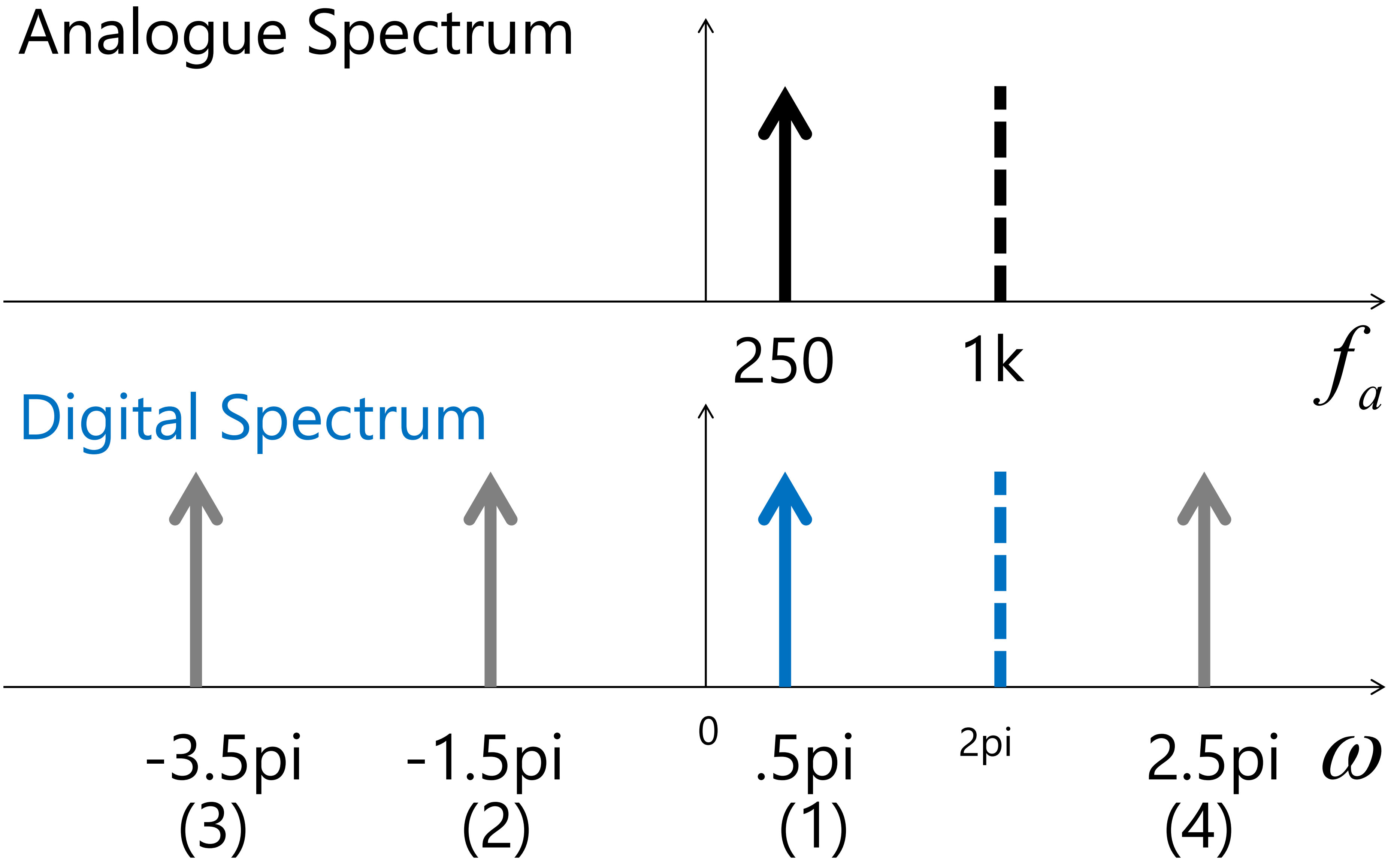
Discrete-time Signals with 2*pi radians apart are the same signals

- If we have x1(n) and x2(n), both are equal when they are 2*pi apart

$$x_1(n) = x_2(n)$$

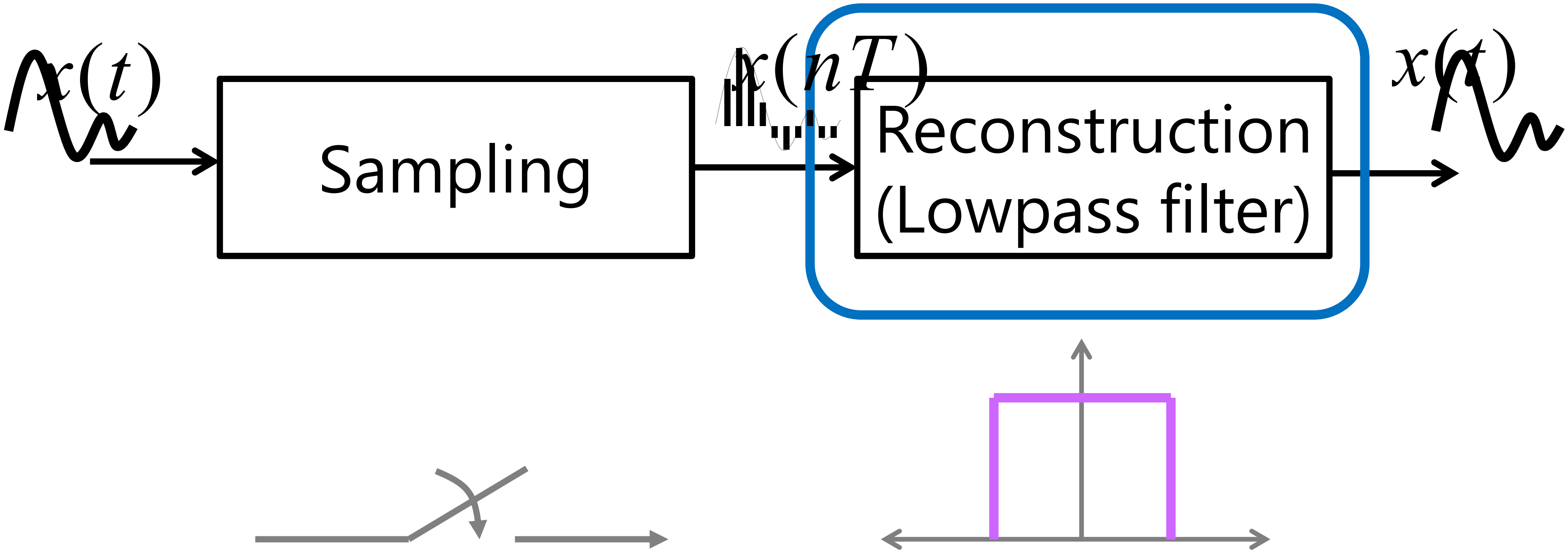
$$\cos(\omega_1 \pi n) = \cos((\omega_1 + 2m)\pi n)$$

$$m = 0, \pm 1, \pm 2, \ldots$$


Analogue Spectrum
250
1k
f_a
Digital Spectrum
-3.5pi
(3)
-1.5pi
(2)
0
.5pi
(1)
2pi
2.5pi
(4)
ω

Reconstruction of the sampled signal

- Convert the digital signal back to analogue



Conversion from digital frequency to analogue frequency

- From the digital frequency,

$$\omega = \Omega T = \Omega \frac{1}{f_s}$$

- So we then have

$$\Omega = \frac{\omega}{T} = \omega f_s$$

- And

$$f_a = \frac{\omega f_s}{2\pi}$$

Cut-off frequency of the reconstruction filter

- We know that the discrete-time signal is discernible only when

$$0 \leq |\omega| \leq \pi$$

- So we set the cut-off frequencies of the ideal reconstruction (lowpass) analogue filter to

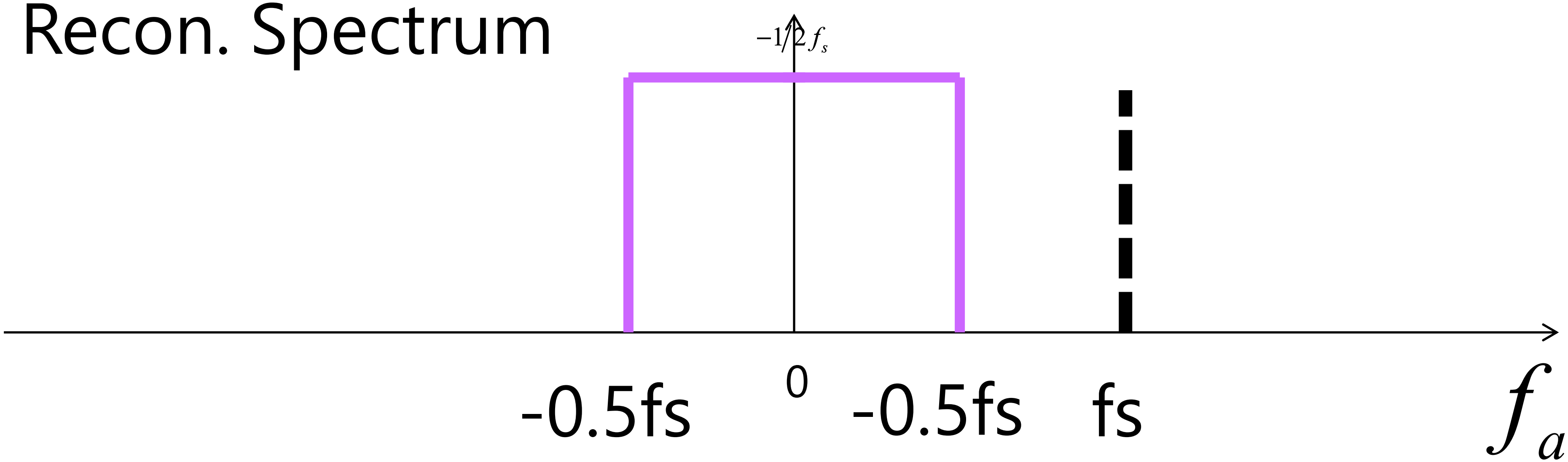
$$-\pi \leq \omega \leq 0 \quad \text{and} \quad 0 \leq \omega \leq \pi$$

$$-\pi \le \omega \le 0 \qquad 0 \le \omega \le \pi$$

$$-\pi \le \Omega / f_s \le 0 \qquad 0 \le \Omega / f_s \le \pi$$

$$-0.5 f_s \le f_a \le 0 \qquad 0 \le f_a \le 0.5 f_s$$

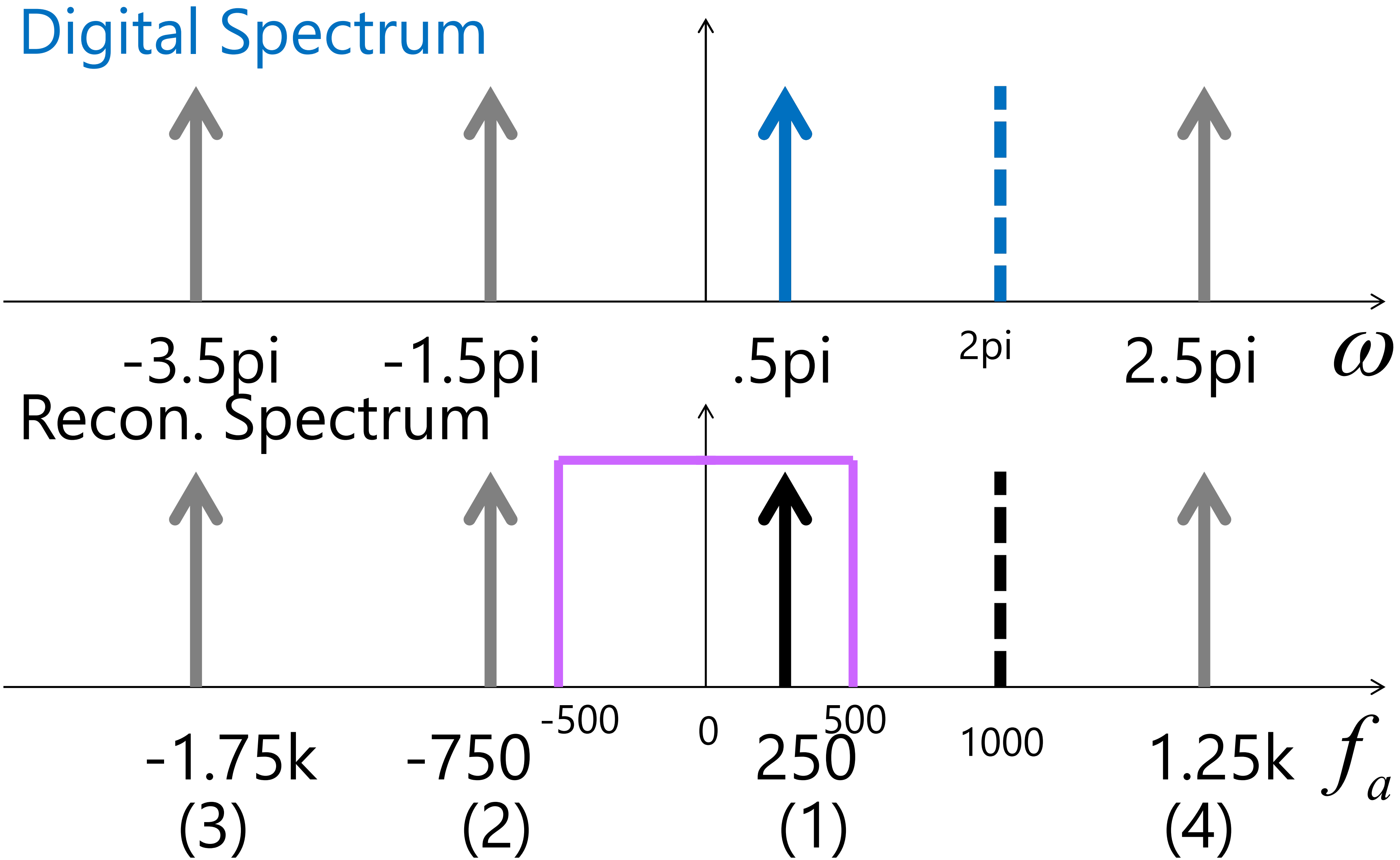
Recon. Spectrum



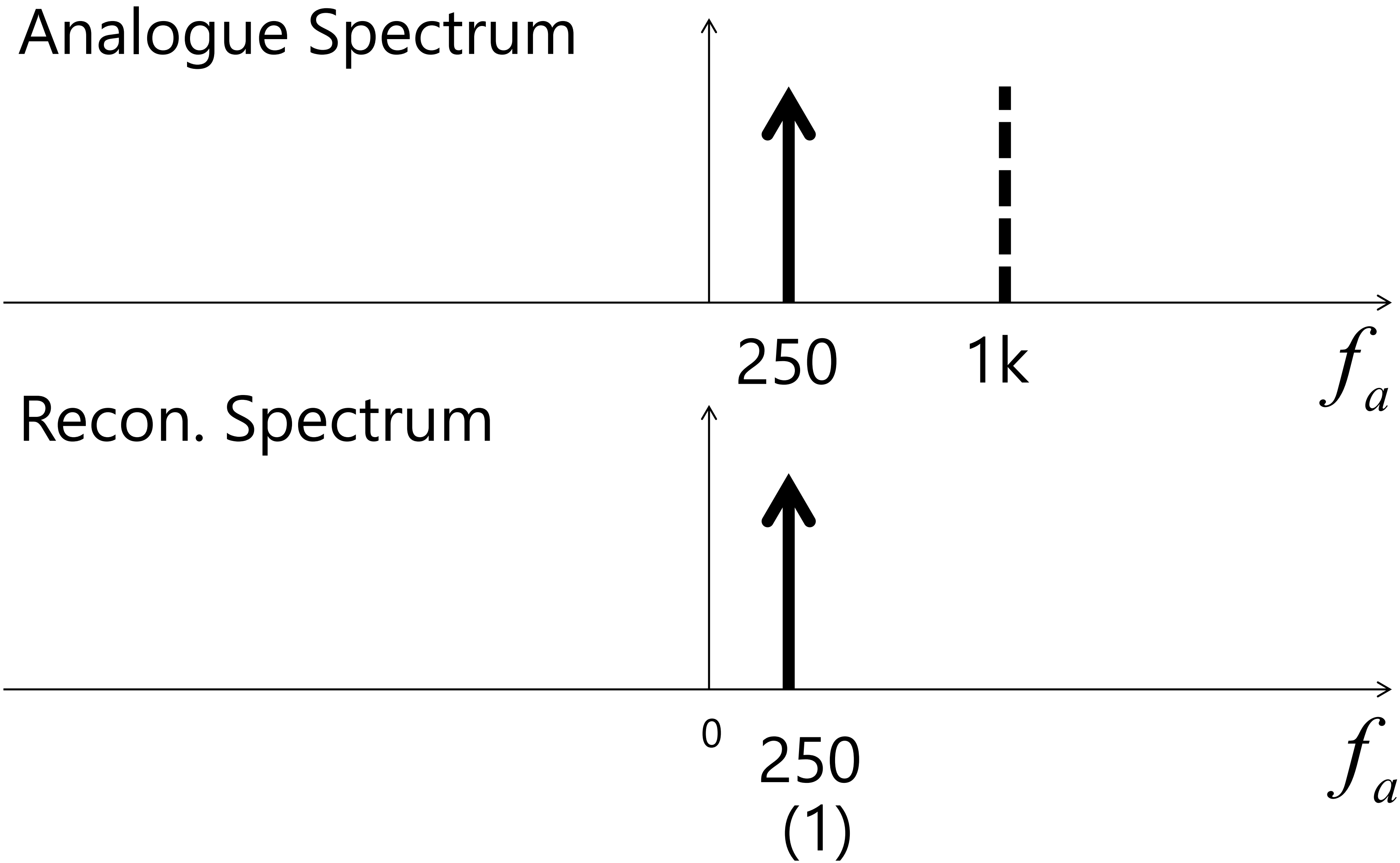
Converting digital frequency to analogue frequency (fs=1,000 Hz)

ω	$\Omega = \omega f_s = \omega(1k)$	$f_a = \omega f_s / 2\pi = \omega(1k)/2\pi$
-3.5π	$-3.5k\pi$	$-1,750$
-1.5π	$-1.5k\pi$	-750
0.5π	500π	250
2.5π	$2.5k\pi$	$1,250$

Perfect Reconstruction



Perfect Reconstruction



Example 3.20: Determine Ambiguity Test

- Suppose we are sampling a sinusoidal signal x(t) with fs =1000 Hz where

$$x(t) = \cos(2500\pi t)$$
$$= \cos(2\pi(1250)t)$$

$$f_a = 1250 \text{ Hz}$$

which means fa is beyond the sampling signal fs.

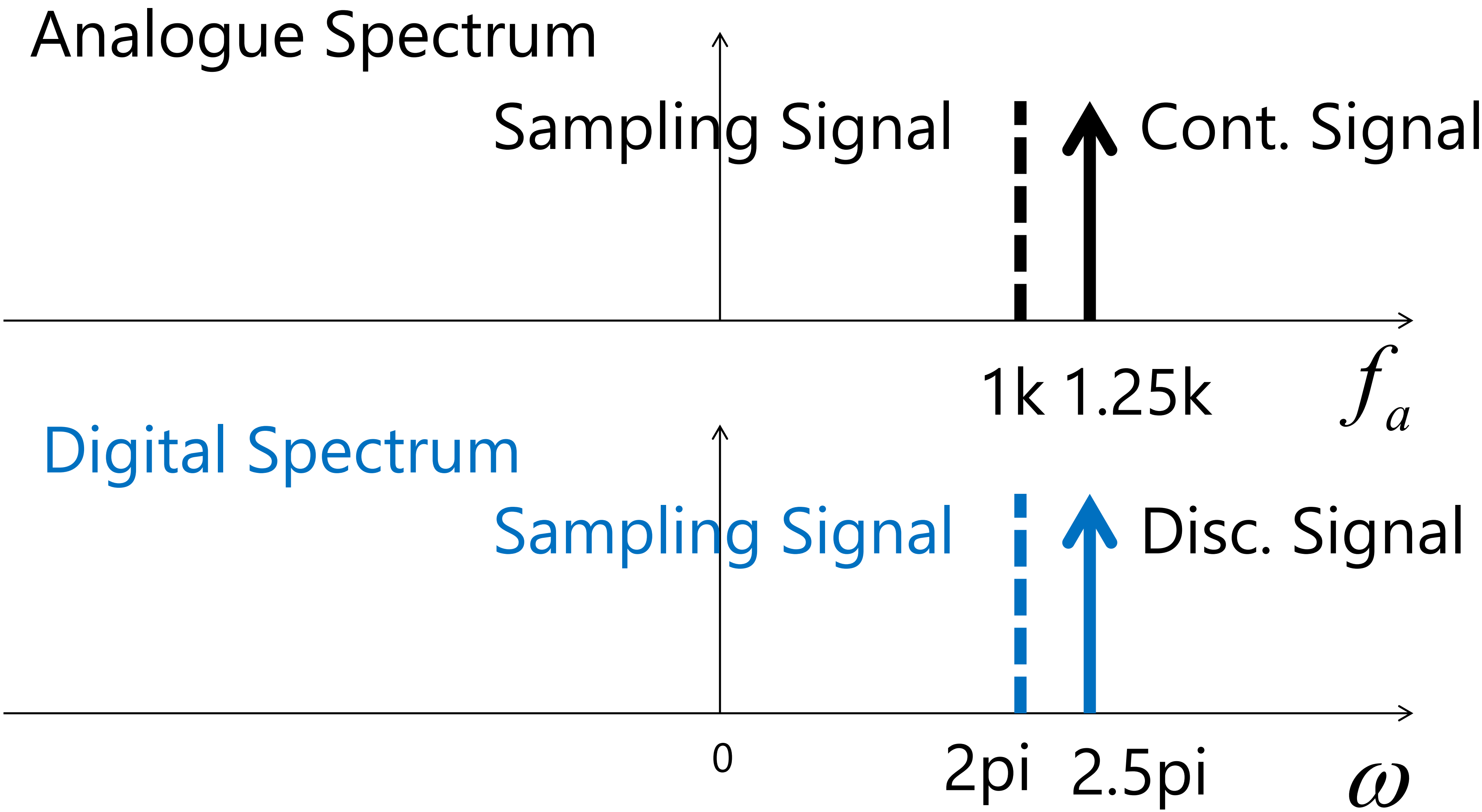
$$(1)\quad x(n)=\cos(2.5\pi n)$$

- From $$\omega=\Omega T=\Omega\frac{1}{f_s}$$

$$\omega=2\pi f_a\frac{1}{f_s}=2500\pi\frac{1}{1000}=2.5\pi$$

- So we have

$$x(n)=\cos(2.5\pi n)$$


Analogue Spectrum
Sampling Signal
Cont. Signal
1k
1.25k
f_a
Digital Spectrum
Sampling Signal
Disc. Signal
0
2pi
2.5pi
ω

(1)

Frequencies Replicas (Copies)

- Again, when sampling one signal in the time domain, we get **more than one frequencies** in the digital frequency domain.
- Let us find three frequency replicas apart from **cos(2.5*pi*n)** produced from sampling a **1,250Hz** x(t) with **fs=1000Hz**.

(2) $x(n) = \cos(0.5\pi n)$

- We can show that

$$
\begin{aligned}
\cos(0.5\pi n) &= \cos((-2+2.5)\pi n) \\
&= \cos(-2\pi n + 2.5\pi n) \\
&= \cos(-2\pi n)\cos(2.5\pi n) \\
&\quad - \sin(-2\pi n)\sin(2.5\pi n)
\end{aligned}
$$

$$\therefore \cos(0.5\pi n) = \cos(2.5\pi n)$$

$$(3)\, x(n) = \cos(-1.5\pi n)$$

- We can show that

$$\begin{aligned}\cos(-1.5\pi n) &= \cos((-4+2.5)\pi n)\\ &= \cos(-4\pi n + 2.5\pi n)\\ &= \cos(-4\pi n)\cos(2.5\pi n)\\ &\quad - \sin(-4\pi n)\sin(2.5\pi n)\end{aligned}$$

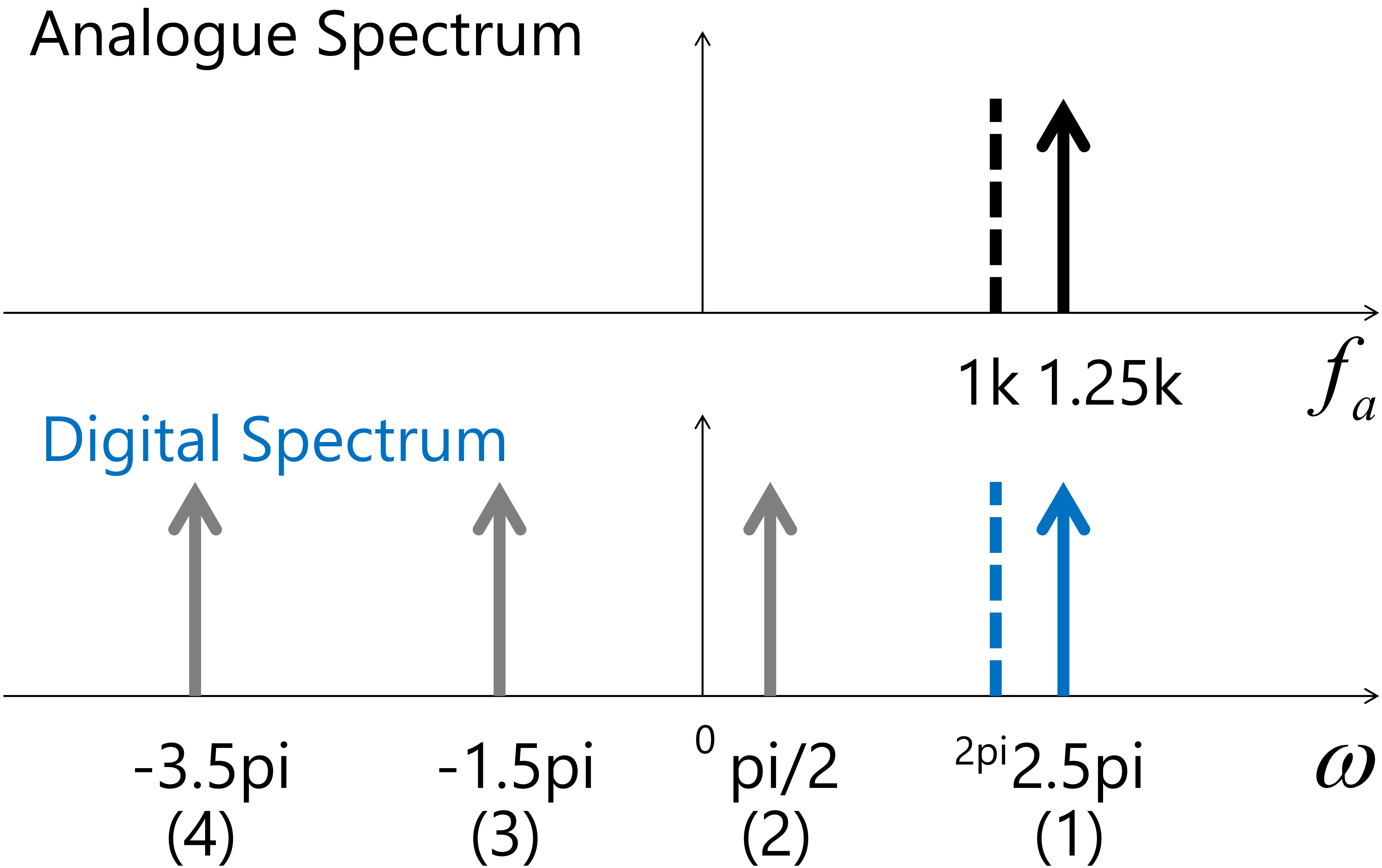
$$\therefore \cos(-1.5\pi n) = \cos(2.5\pi n)$$

$$(4)\, x(n) = \cos(-1.5\pi n)$$

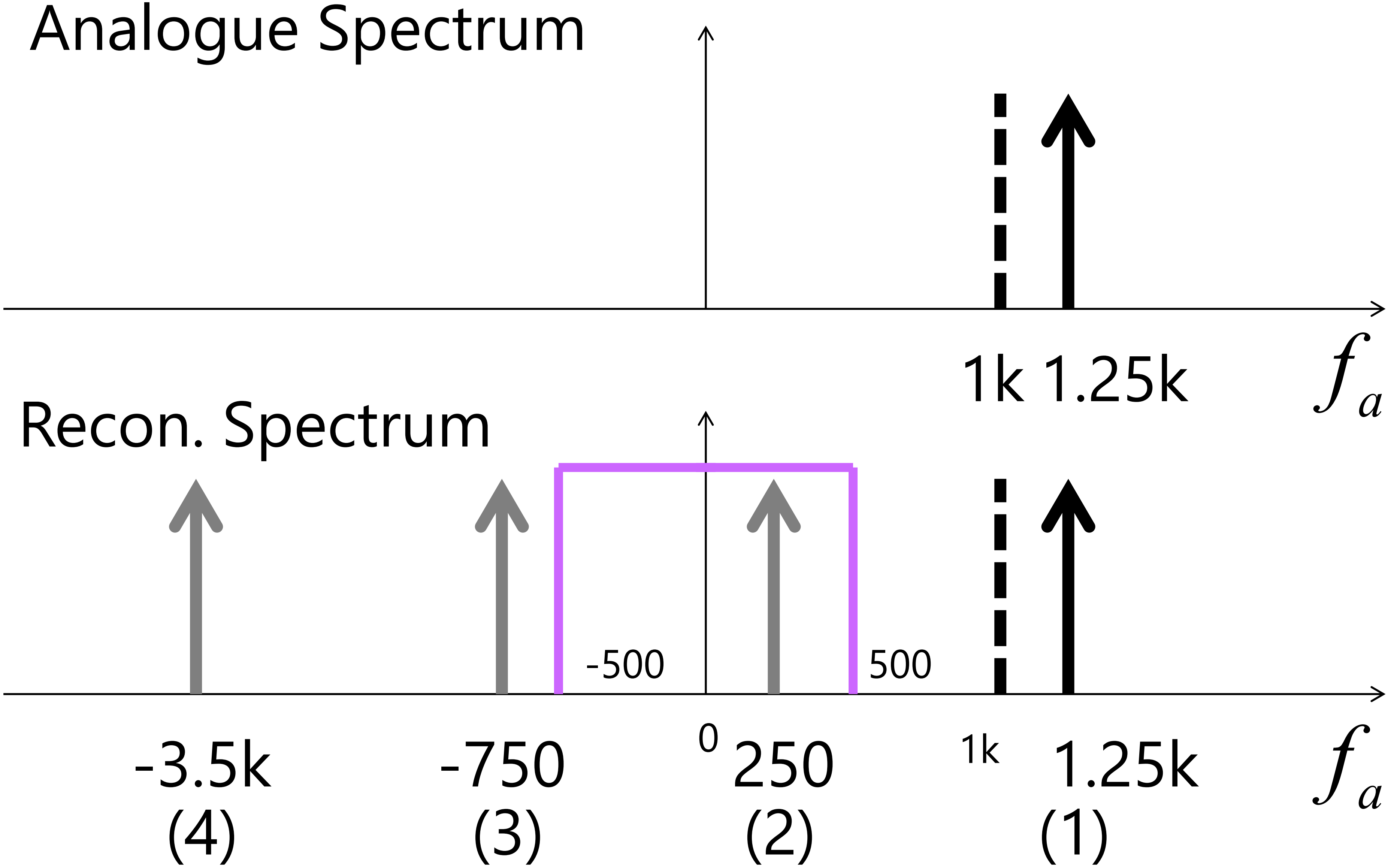
- We can show that

$$\cos(-3.5\pi n) = \cos((-6+2.5)\pi n)$$
$$= \cos(-4\pi n + 2.5\pi n)$$
$$= \cos(-6\pi n)\cos(2.5\pi n)$$
$$- \sin(-6\pi n)\sin(2.5\pi n)$$

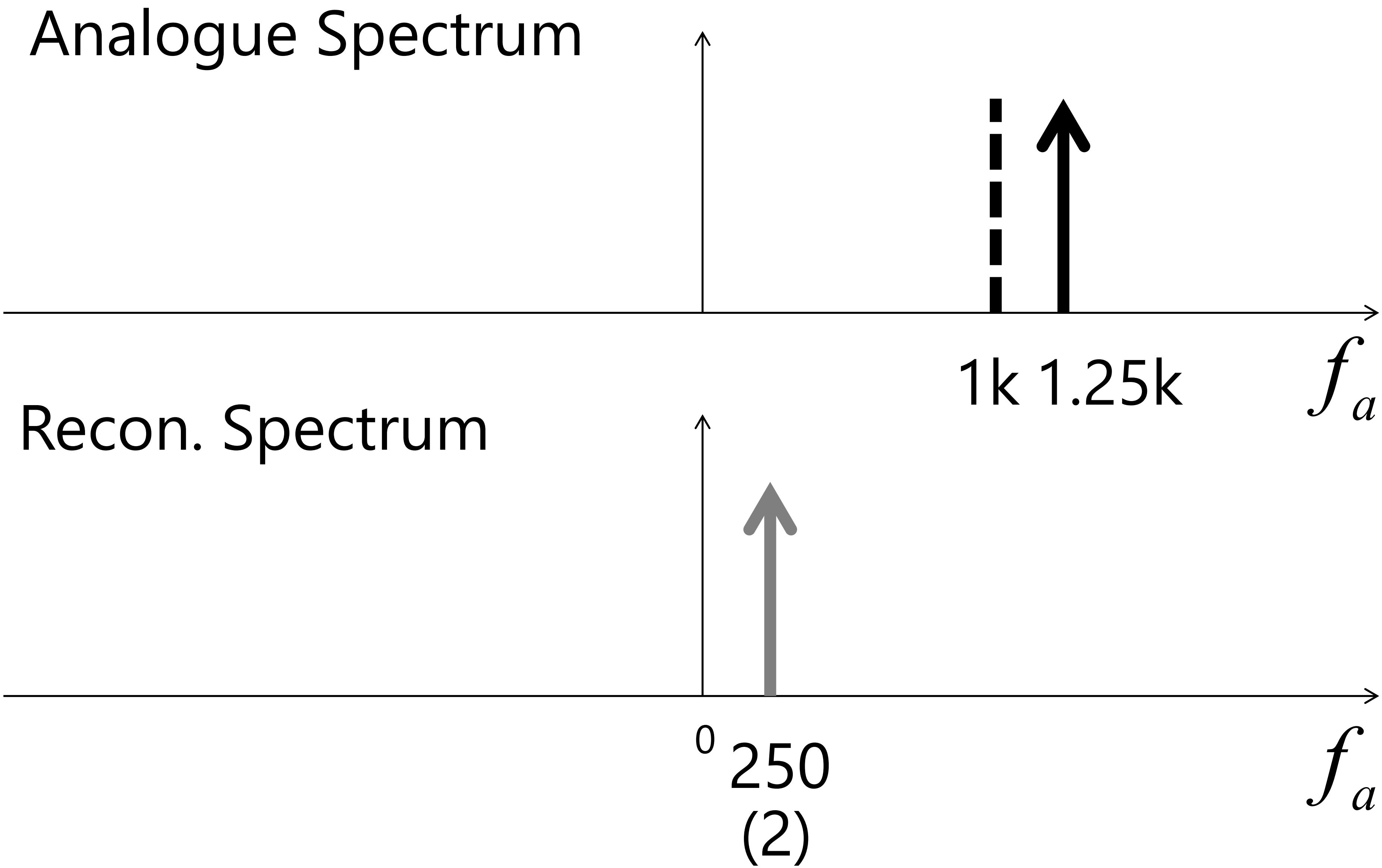
$$\therefore \cos(-3.5\pi n) = \cos(2.5\pi n)$$


Analogue Spectrum
1k 1.25k
f_a
Digital Spectrum
-3.5pi
(4)
-1.5pi
(3)
0
pi/2
(2)
2pi
2.5pi
(1)
ω

Incorrect Reconstruction



Incorrect Reconstruction



Example 3.20: Determine Ambiguity Test

- Suppose we are sampling a sinusoidal signal x(t) with fs =1000 Hz where

$$x(t) = \cos(1000\pi t)$$
$$= \cos(2\pi(500)t)$$

$$f_a = 500 \text{ Hz}$$

which means fa is **NOT** beyond the sampling signal fs.

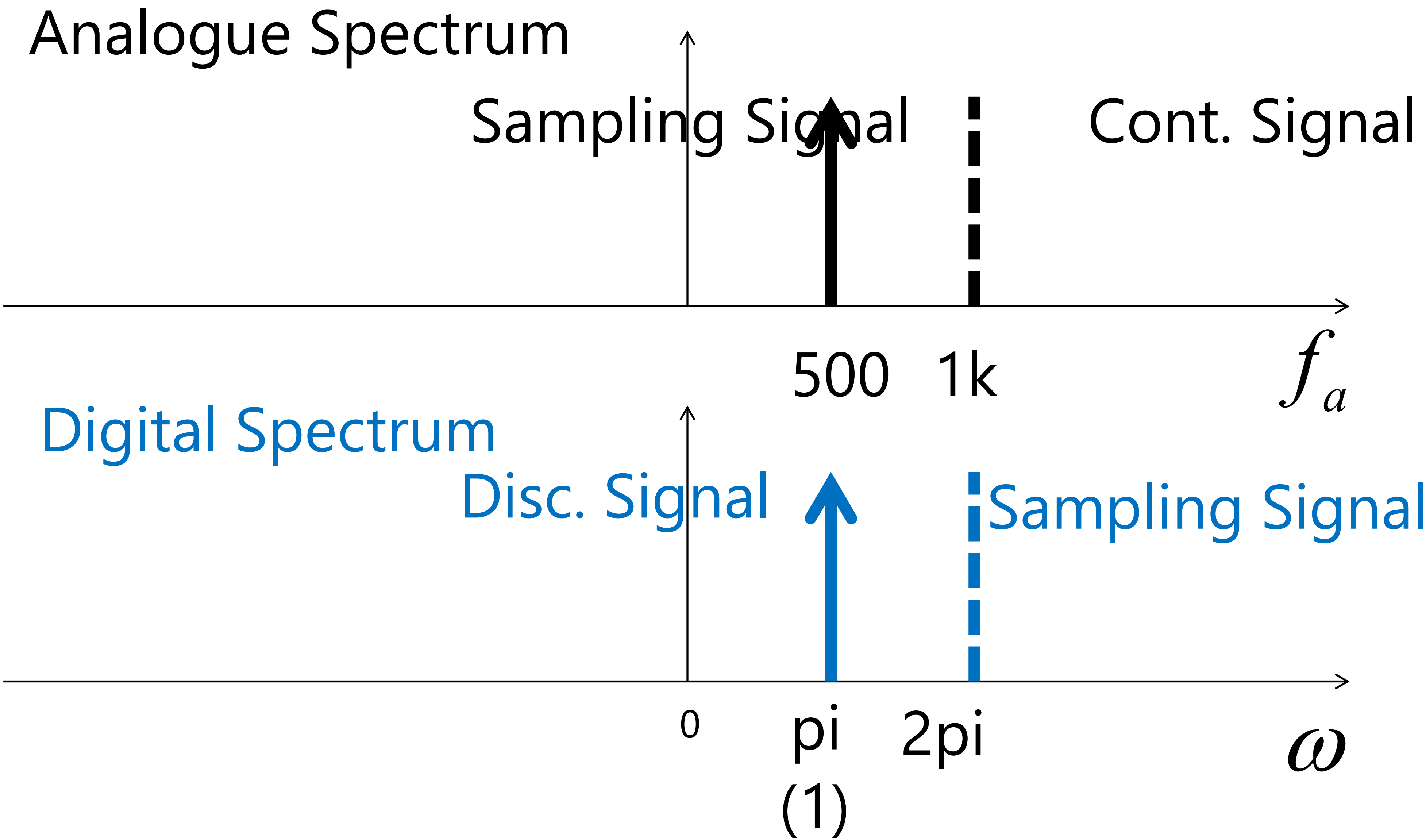
(1) $x(n) = \cos(2.5\pi n)$

- From $$\omega = \Omega T = \Omega \frac{1}{f_s}$$

$$\omega = 2\pi f_a \frac{1}{f_s} = 1000\pi \frac{1}{1000} = \pi$$

- So we have

$$x(n) = \cos(\pi n)$$


Analogue Spectrum
Sampling Signal
Cont. Signal
500
1k
f_a
Digital Spectrum
Disc. Signal
Sampling Signal
0
pi
(1)
2pi
ω

- However, x(n) with digital frequency pi radians is not only the signal we can produce from sampling the 500Hz x(t) with fs=1000Hz.
- Another frequencies can be produced too!
- Let us try

$$(2) \quad x(n) = \cos(-\pi n)$$

- We can show that

$$\begin{aligned}
\cos(-\pi n) &= \cos((-2+1)\pi n) \\
&= \cos(-2\pi n + \pi n) \\
&= \cos(-2\pi n)\cos(\pi n) \\
&\quad - \sin(-2\pi n)\sin(\pi n)
\end{aligned}$$

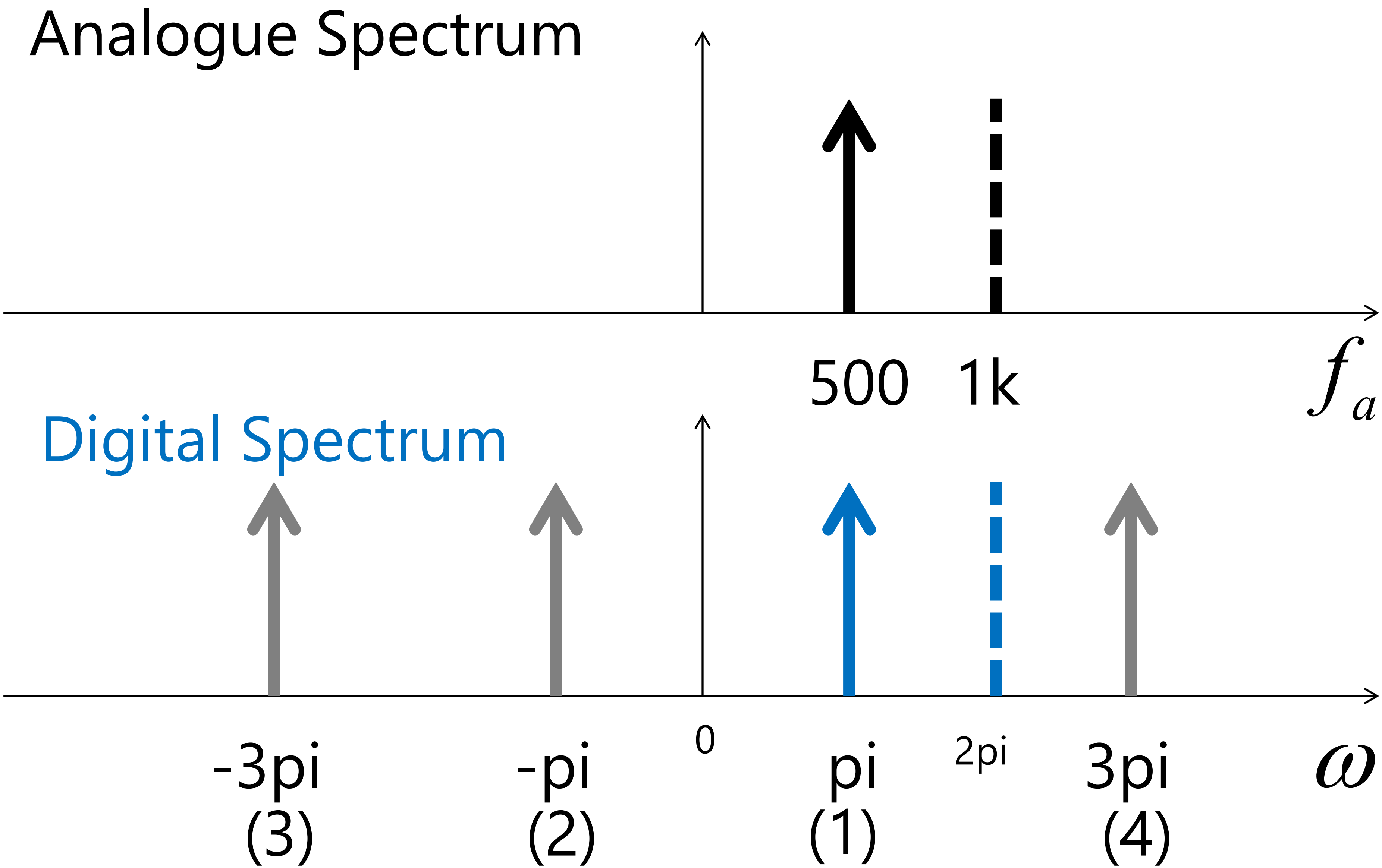
$$\therefore \cos(-\pi n) = \cos(\pi n)$$

(3) $x(n) = \cos(-3\pi n)$

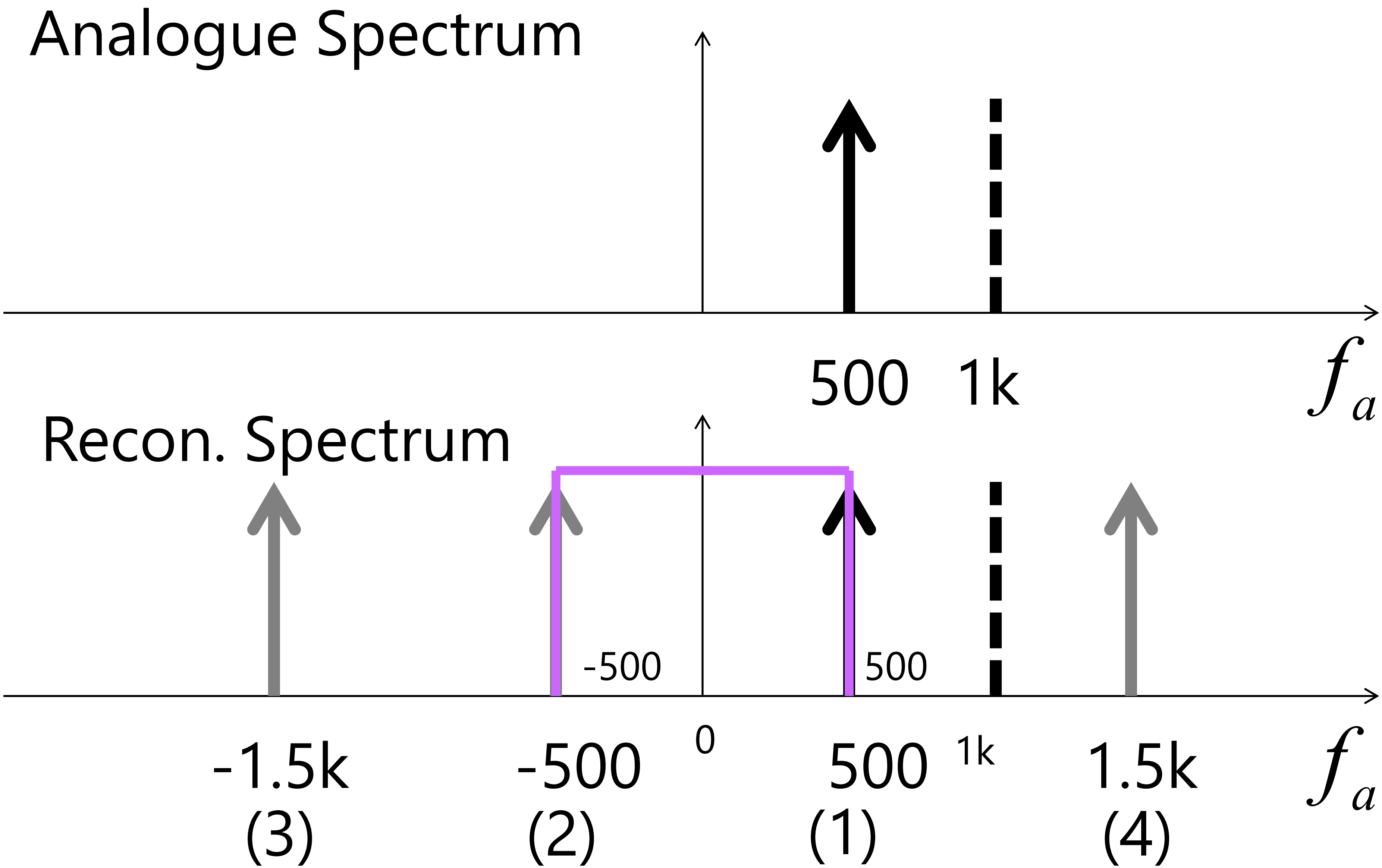
- We can show that

$$\begin{aligned}\cos(-3\pi n) = \cos((-4+2.5)\pi n) &= \cos(-4\pi n + \pi n) \\ &= \cos(-4\pi n)\cos(\pi n) \\ &\quad - \sin(-4\pi n)\sin(\pi n)\end{aligned}$$

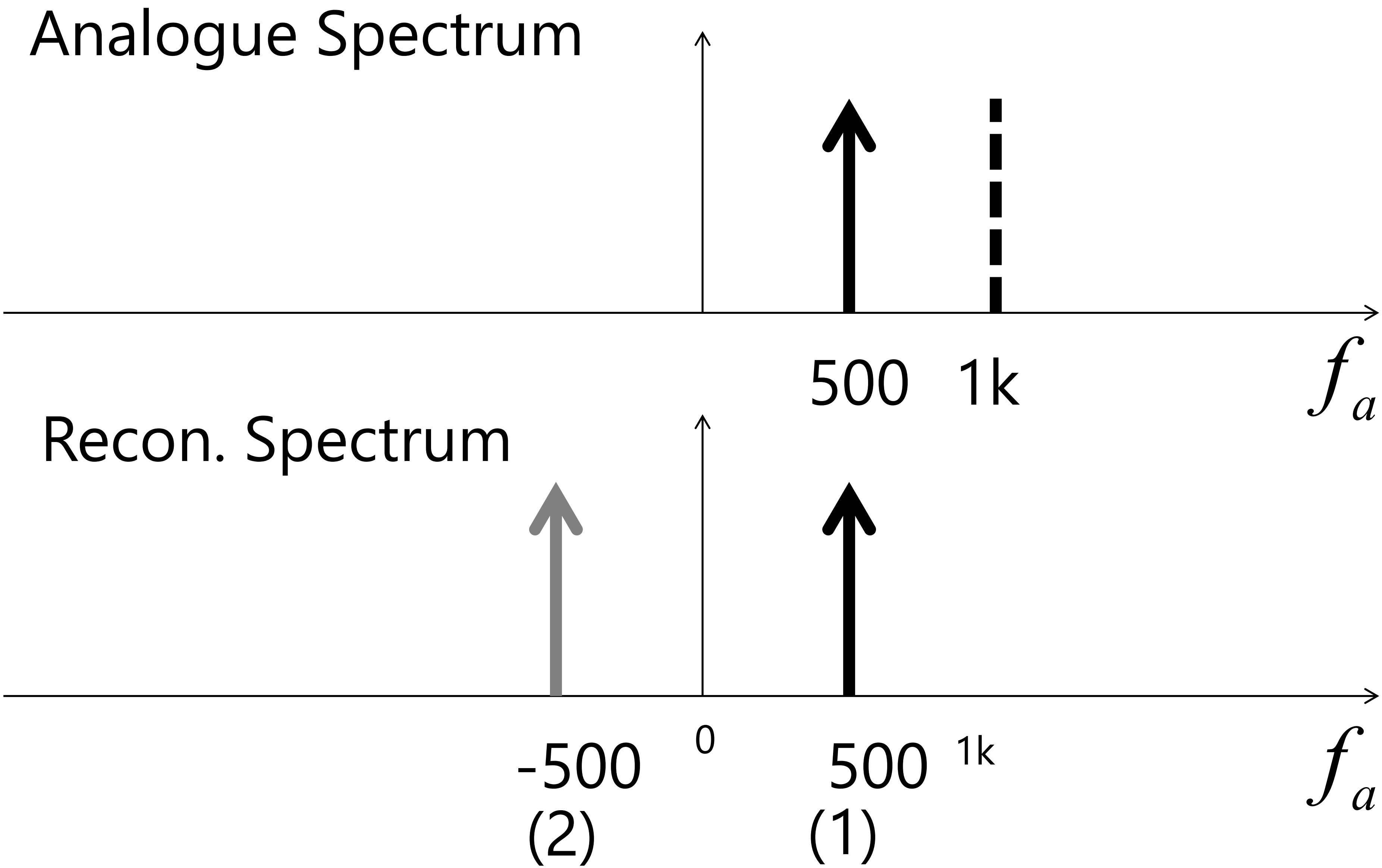
$$\therefore \cos(-3\pi n) = \cos(\pi n)$$


Analogue Spectrum
500
1k
f_a
Digital Spectrum
-3pi
(3)
-pi
(2)
0
pi
(1)
2pi
3pi
(4)
ω

Incorrect Reconstruction



Incorrect Reconstruction



Example 3.21: Determine Ambiguity Test

- Suppose we are sampling a sinusoidal signal x(t) with has frequency components of signals

$$x(t) = A\cos(\Omega t) \qquad -1000\pi \le \Omega \le 1000\pi$$

$$x_{\min}(t) = A_{\min}\cos(-1000\pi t) = A_{\min}\cos(2\pi(-500)t)$$

$$x_{\max}(t) = A_{\max}\cos(1000\pi t) = A_{\max}\cos(2\pi(500)t)$$

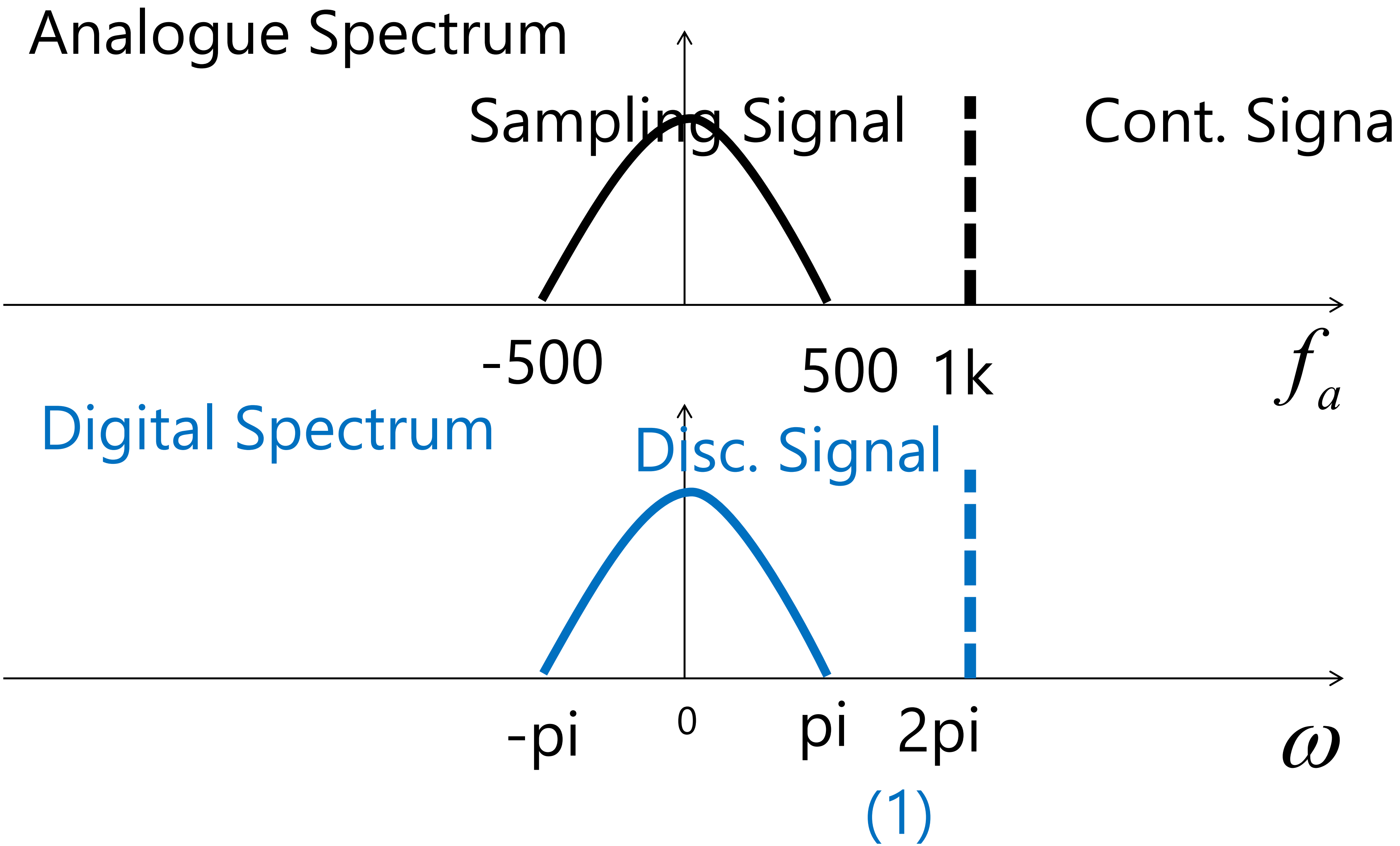
(1)

- From $$\omega = \Omega T = \Omega \frac{1}{f_s}$$

$$\omega_{\min} = 2\pi f_{a,\min} \frac{1}{f_s} = -1000\pi \frac{1}{1000} = -\pi$$

$$\omega_{\max} = 2\pi f_{a,\max} \frac{1}{f_s} = 100\pi \frac{1}{1000} = \pi$$

- So we have
$$x_{\min}(n) = A_{\min} \cos(-\pi n)$$
$$x_{\max}(n) = A_{\max} \cos(\pi n)$$


Analogue Spectrum
Sampling Signal
Cont. Signal
-500
500
1k
f_a
Digital Spectrum
Disc. Signal
-pi
0
pi
2pi
ω
(1)

$$x_{\min}(n) = \cos(-1.5\pi n), x_{\max}(n) = \cos(0.5\pi n)$$

- We can show that

$$\cos(-3\pi n) = \cos((-2 + -1)\pi n)$$

$$= \cos(-2\pi n + -\pi n)$$

$$= \cos(-2\pi n)\cos(-\pi n)$$

$$- \sin(-2\pi n)\sin(-\pi n)$$

$$\therefore \cos(-3\pi n) = \cos(-\pi n)$$

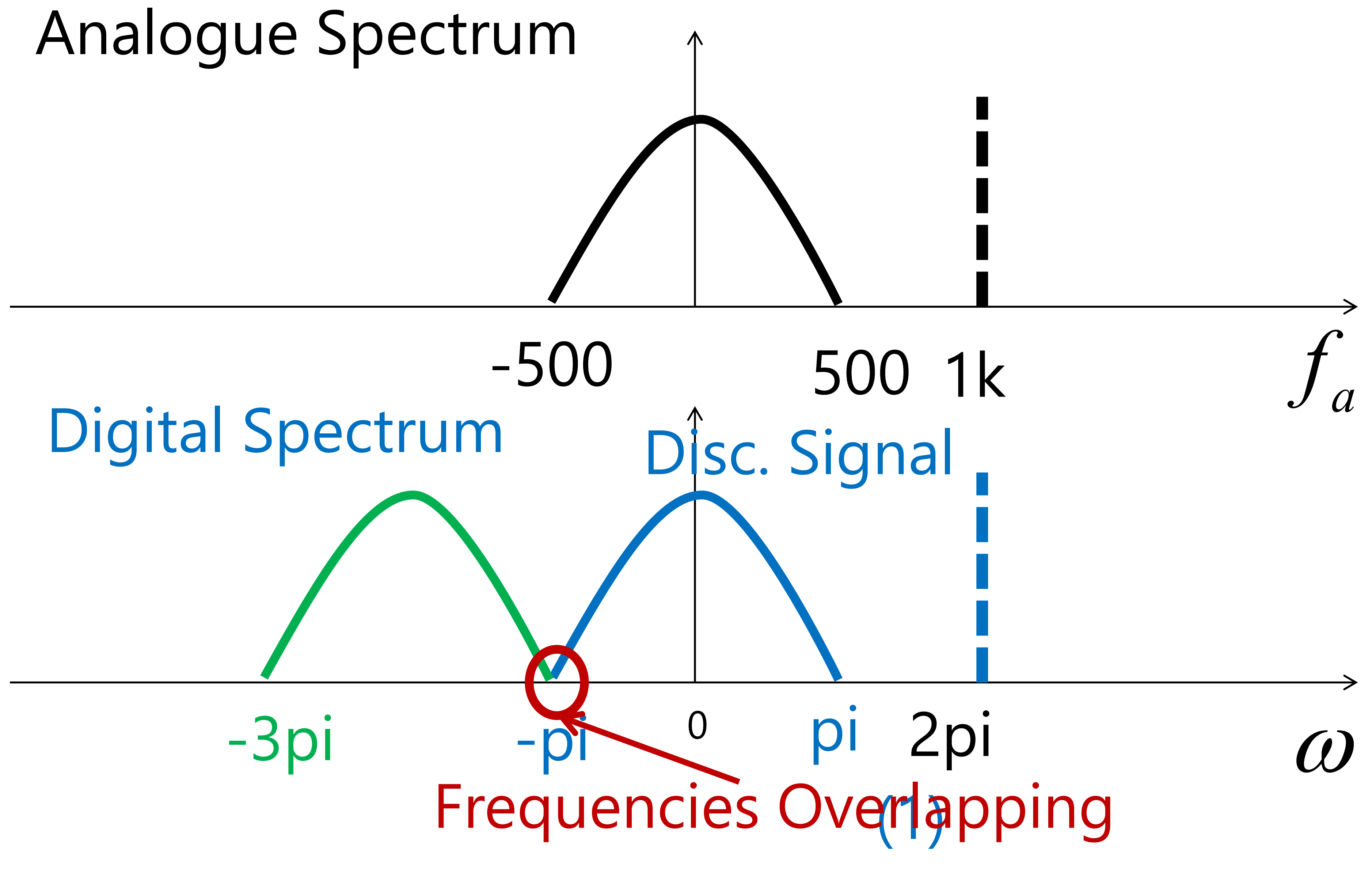
$$\cos(-\pi n) = \cos((-2 + 1)\pi n)$$

$$= \cos(-2\pi n + \pi n)$$

$$= \cos(-2\pi n)\cos(\pi n)$$

$$- \sin(-2\pi n)\sin(\pi n)$$

$$\therefore \cos(-\pi n) = \cos(\pi n)$$



Analogue Spectrum
-500
500
1k
f_a
Digital Spectrum
Disc. Signal
-3pi
-pi
0
pi
2pi
ω
Frequencies Overlapping
(1)

(2) $x(n) = \cos(-1.5\pi n)$

- We can show that

$$\cos(\pi n) = \cos((2 + -1)\pi n)$$

$$= \cos(2\pi n - \pi n)$$

$$= \cos(2\pi n)\cos(\pi n)$$

$$- \sin(2\pi n)\sin(\pi n)$$

$$\therefore \cos(\pi n) = \cos(-\pi n)$$

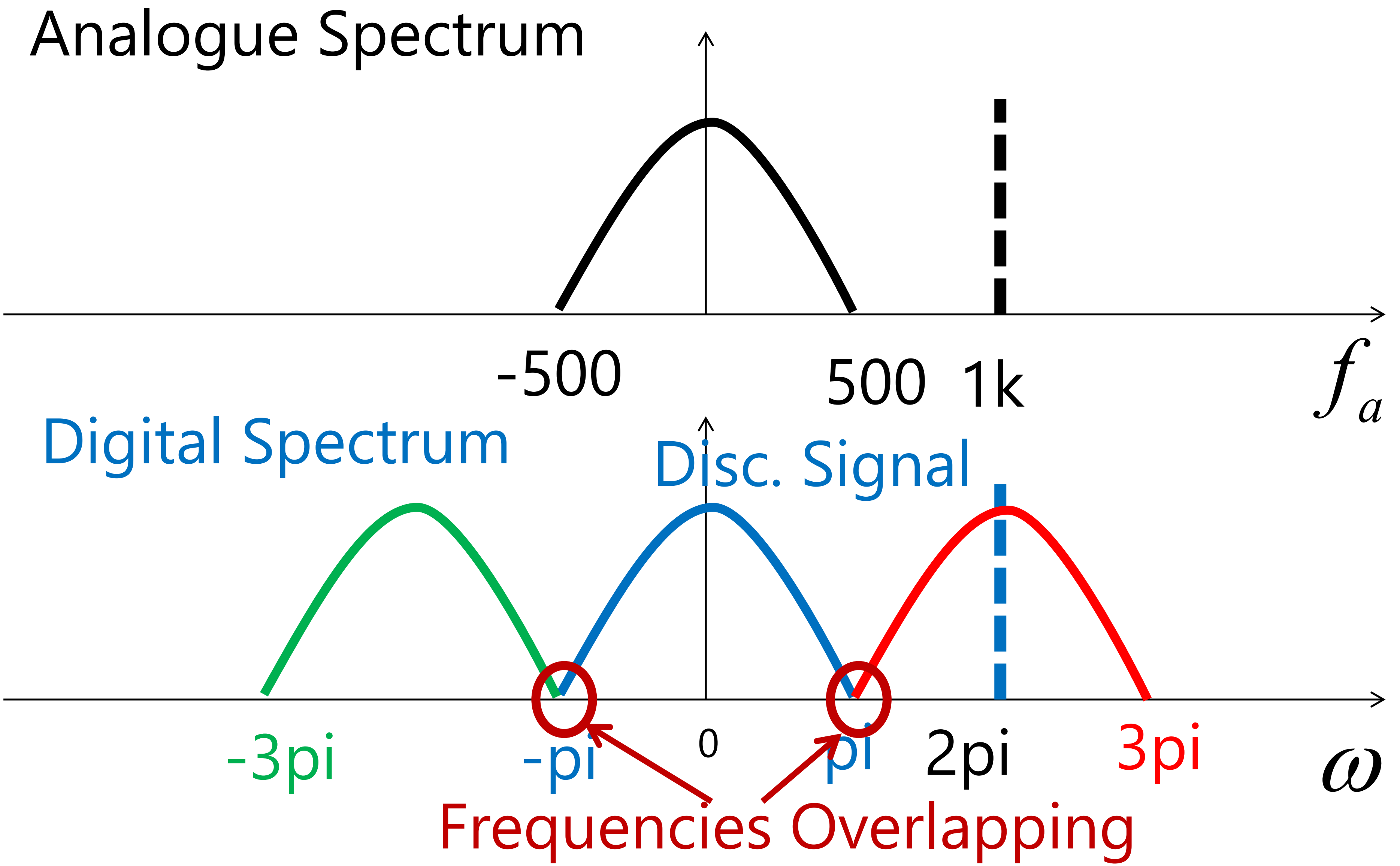
$$\cos(3\pi n) = \cos((2 + 1)\pi n)$$

$$= \cos(2\pi n + \pi n)$$

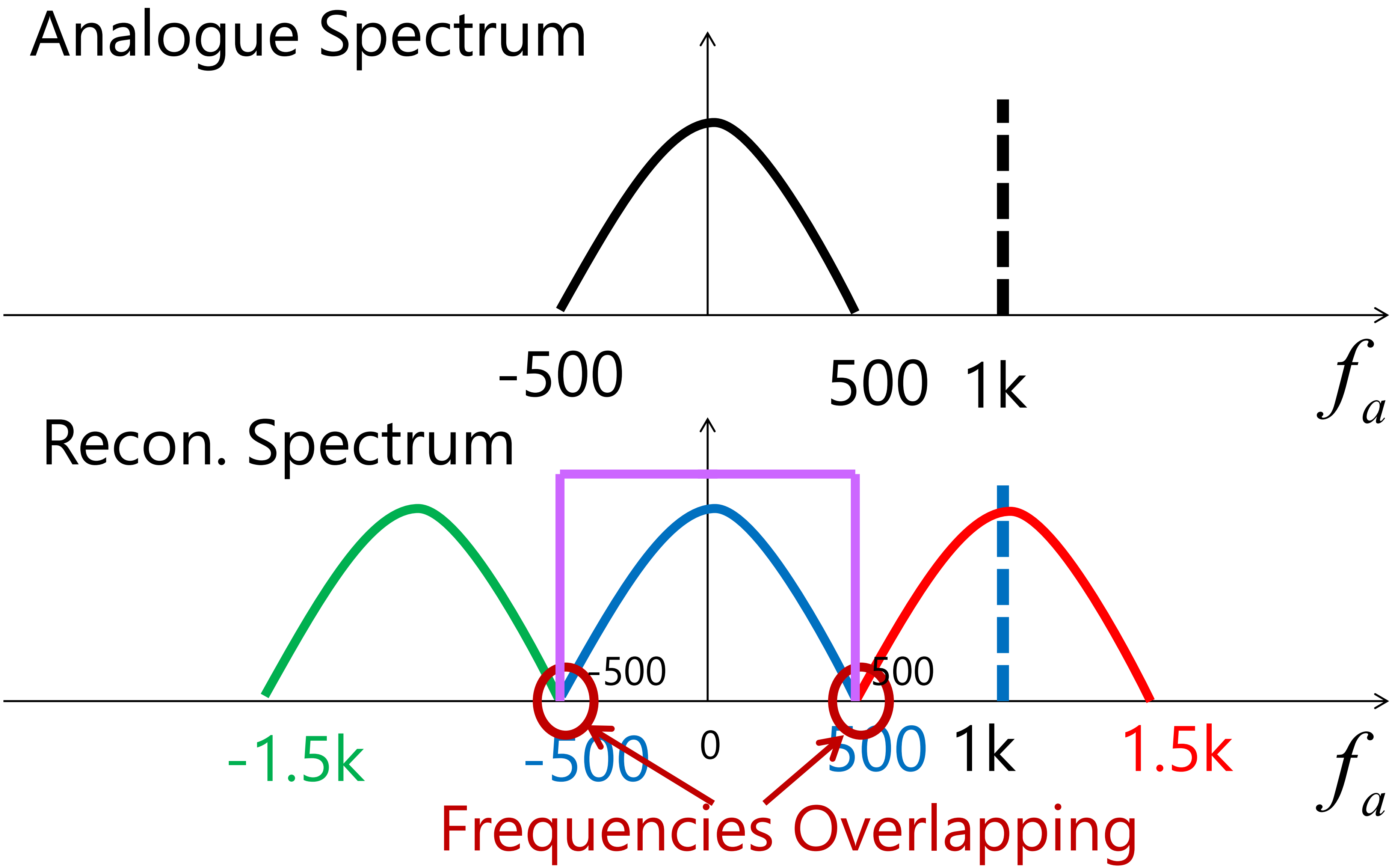
$$= \cos(2\pi n)\cos(\pi n)$$

$$- \sin(2\pi n)\sin(\pi n)$$

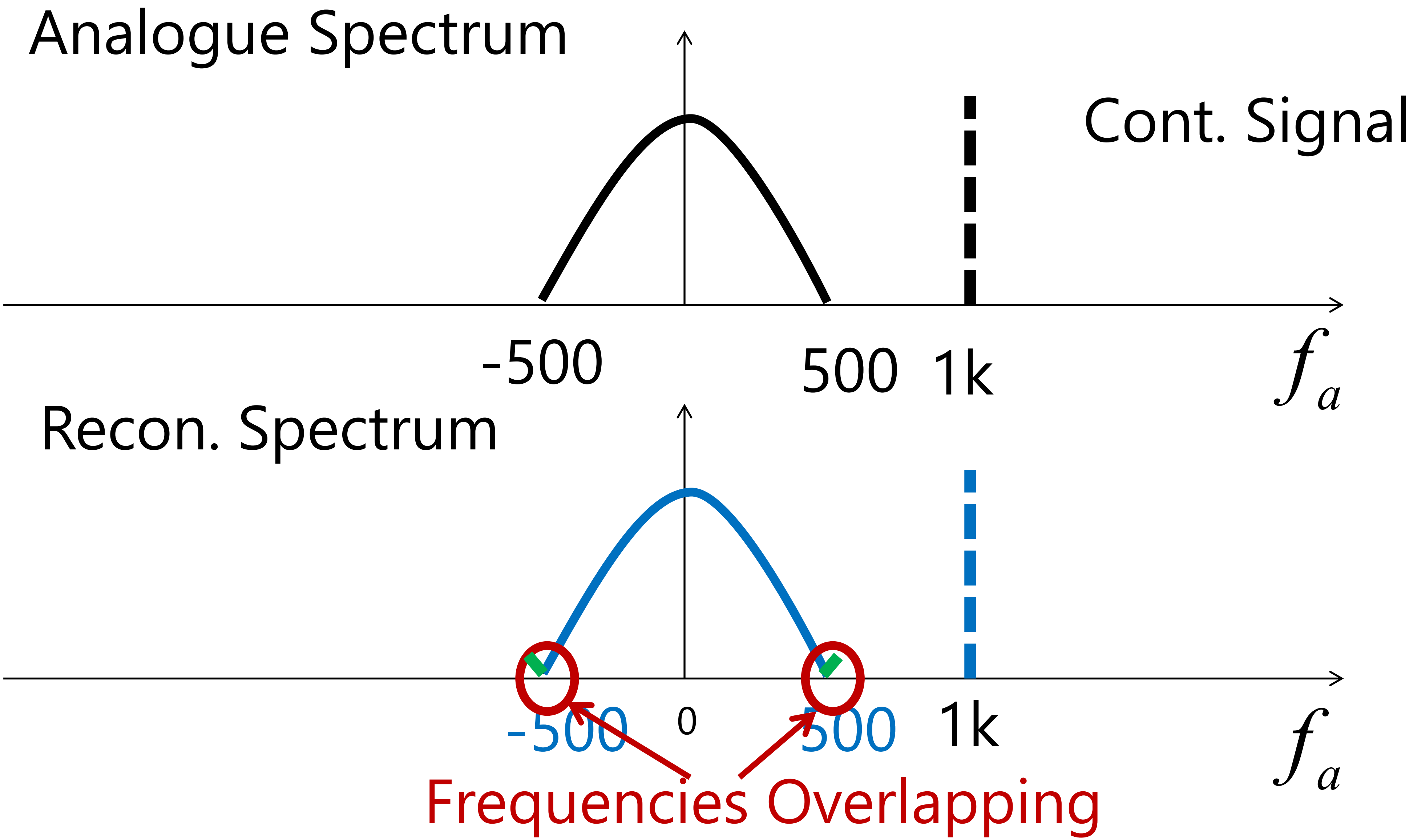
$$\therefore \cos(3\pi n) = \cos(\pi n)$$


Analogue Spectrum
-500
500
1k
f_a
Digital Spectrum
Disc. Signal
-3pi
-pi
0
pi
2pi
3pi
ω
Frequencies Overlapping

Reconstruction with Alias



Reconstruction with Alias



Example 3.22: Determine Ambiguity Test

- Suppose we are sampling a sinusoidal signal x(t) with has frequency components of signals

$$x(t) = A\cos(\Omega t) \qquad -1500\pi \le \Omega \le 1500\pi$$

$$x_{\min}(t) = A_{\min}\cos(2\pi(-750)t)$$

$$x_{\max}(t) = A_{\max}\cos(2\pi(750)t)$$

which means fa is the sampling signal fs.

$$(1)\quad x(n) = \cos(2.5\pi n)$$

- From $$\omega = \Omega T = \Omega \frac{1}{f_s}$$

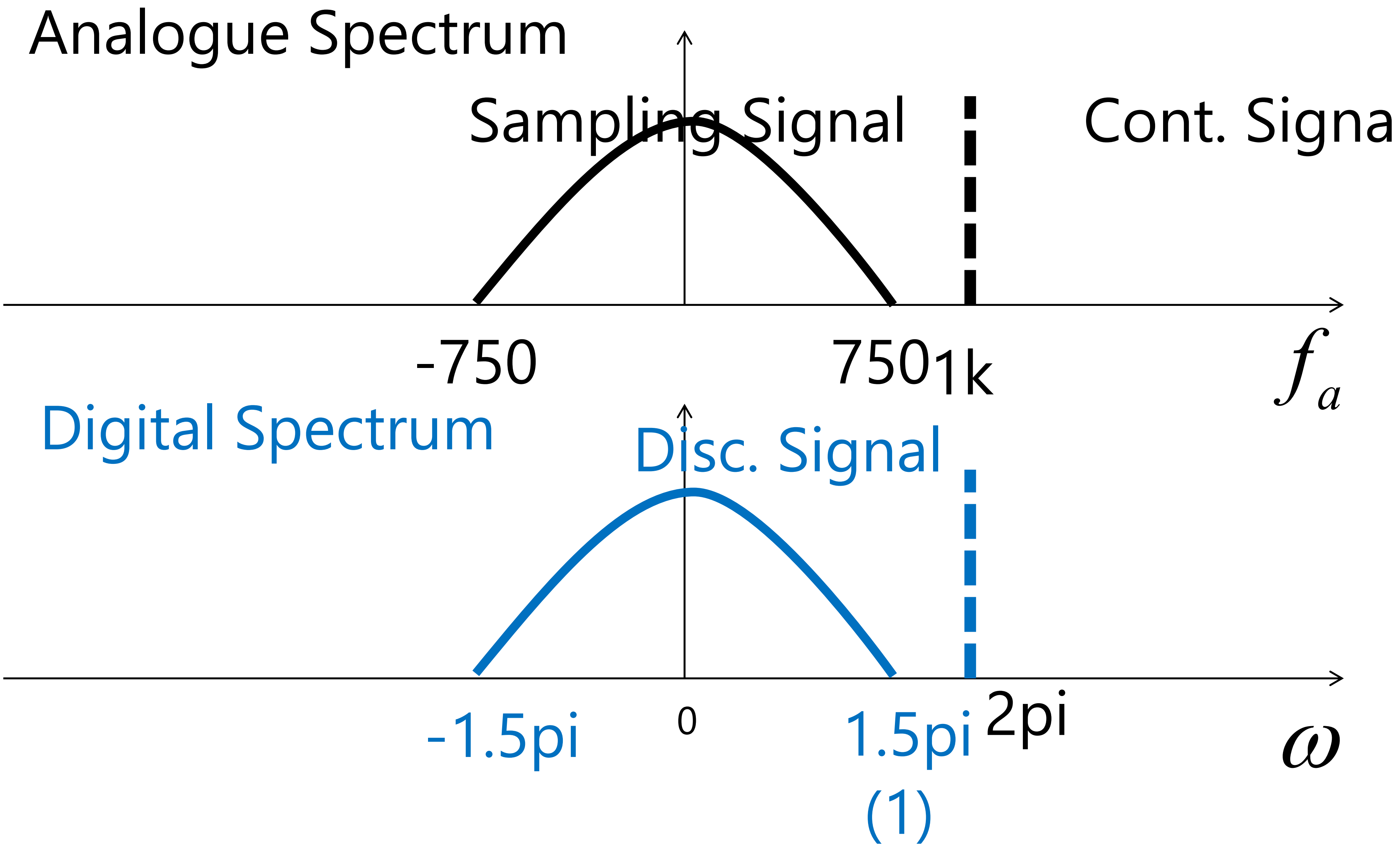
$$\omega_{\min} = 2\pi f_{a,\min} \frac{1}{f_s} = -1500\pi \frac{1}{1000} = -1.5\pi$$

$$\omega_{\max} = 2\pi f_{a,\max} \frac{1}{f_s} = 1500\pi \frac{1}{1000} = 1.5\pi$$

- So we have

$$x_{\min}(n) = A_{\min} \cos(-1.5\pi n)$$

$$x_{\max}(n) = A_{\max} \cos(1.5\pi n)$$



Analogue Spectrum
Sampling Signal
Cont. Signal
-750
750
1k
f_a
Digital Spectrum
Disc. Signal
-1.5pi
0
1.5pi
2pi
ω
(1)

$$x_{\min}(n)=\cos(-1.5\pi n),x_{\max}(n)=\cos(0.5\pi n)$$

- We can show that

$$\cos(-3.5\pi n)=\cos((-2-1.5)\pi n)$$

$$=\cos(-2\pi n+-1.5\pi n)$$

$$=\cos(-2\pi n)\cos(-1.5\pi n)$$

$$-\sin(-2\pi n)\sin(-1.5\pi n)$$

$$\therefore\cos(-3.5\pi n)=\cos(-1.5\pi n)$$

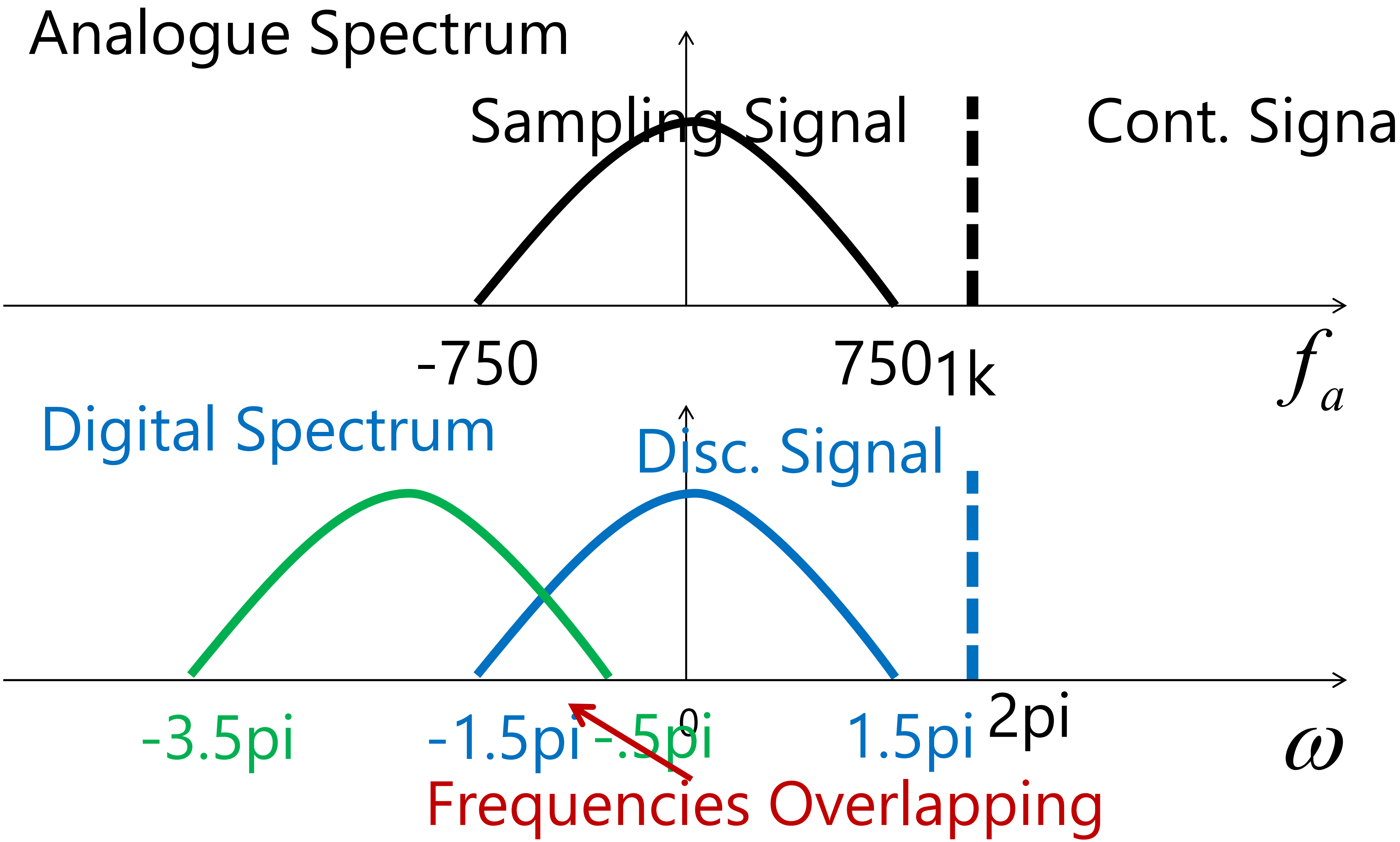
$$\cos(-0.5\pi n)=\cos((-2+1.5)\pi n)$$

$$=\cos(-2\pi n+1.5\pi n)$$

$$=\cos(-2\pi n)\cos(1.5\pi n)$$

$$-\sin(-2\pi n)\sin(1.5\pi n)$$

$$\therefore\cos(-0.5\pi n)=\cos(1.5\pi n)$$


Analogue Spectrum
Sampling Signal
Cont. Signal
-750
750
1k
f_a
Digital Spectrum
Disc. Signal
-3.5pi
-1.5pi
-.5pi
0
1.5pi
2pi
ω
Frequencies Overlapping

$$x_{\min}(n) = \cos(-1.5\pi n), x_{\max}(n) = \cos(0.5\pi n)$$

- We can show that

$$\cos(.5\pi n) = \cos((2-1.5)\pi n)$$

$$= \cos(2\pi n - 1.5\pi n)$$

$$= \cos(2\pi n)\cos(-1.5\pi n)$$

$$- \sin(2\pi n)\sin(-1.5\pi n)$$

$$\therefore \cos(.5\pi n) = \cos(-1.5\pi n)$$

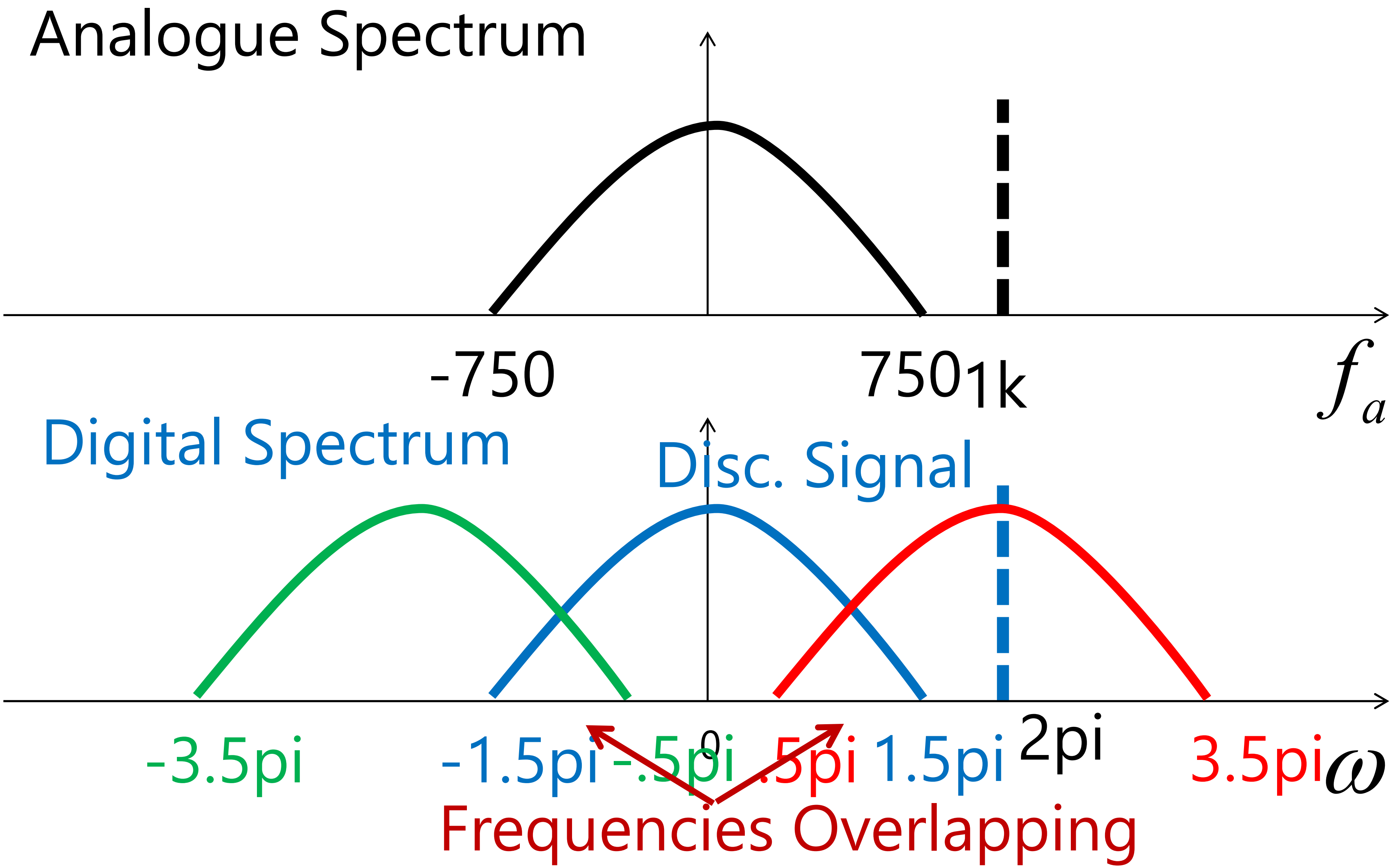
$$\cos(3.5\pi n) = \cos((2+1.5)\pi n)$$

$$= \cos(2\pi n + 1.5\pi n)$$

$$= \cos(2\pi n)\cos(1.5\pi n)$$

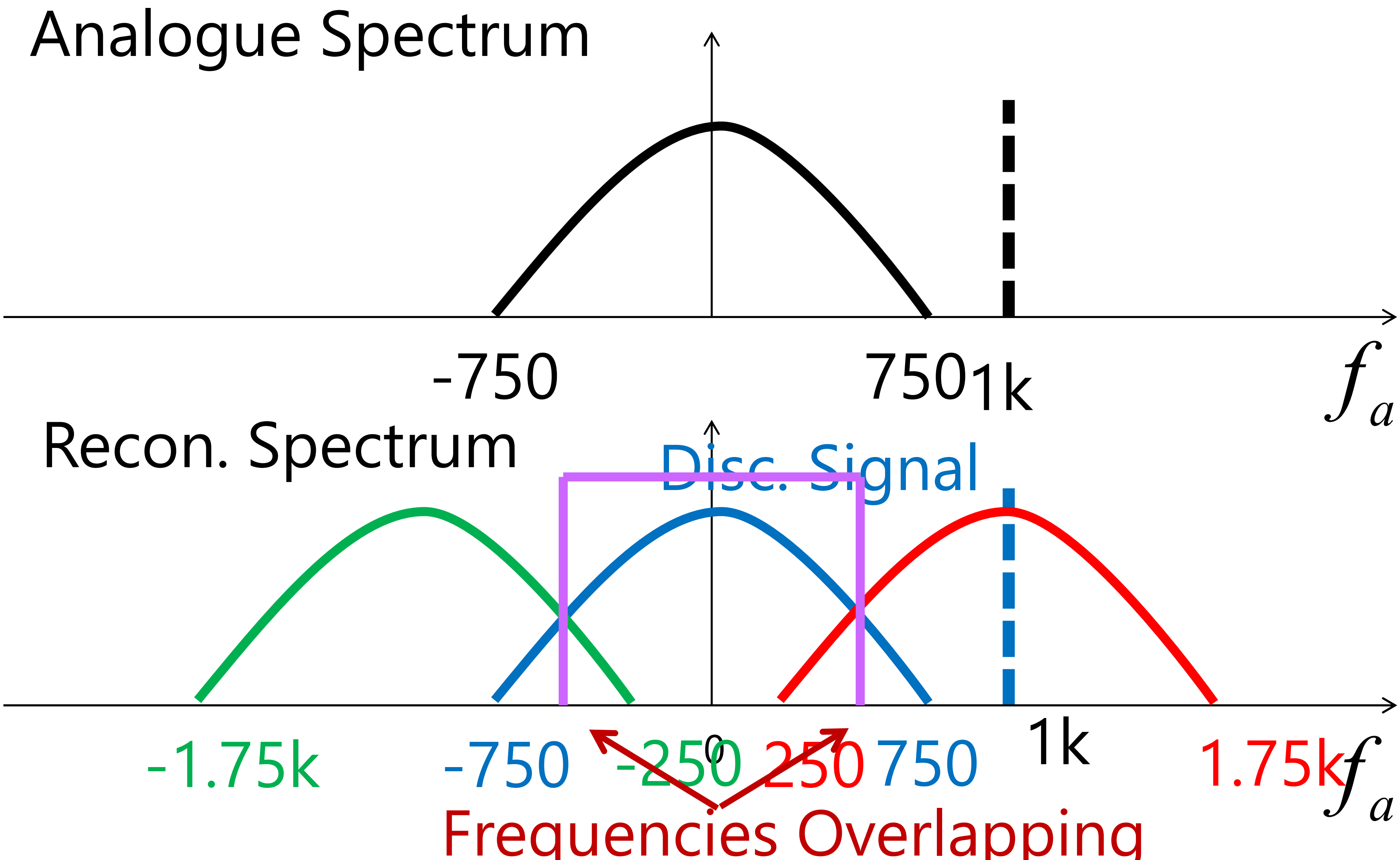
$$- \sin(2\pi n)\sin(1.5\pi n)$$

$$\therefore \cos(3.5\pi n) = \cos(1.5\pi n)$$

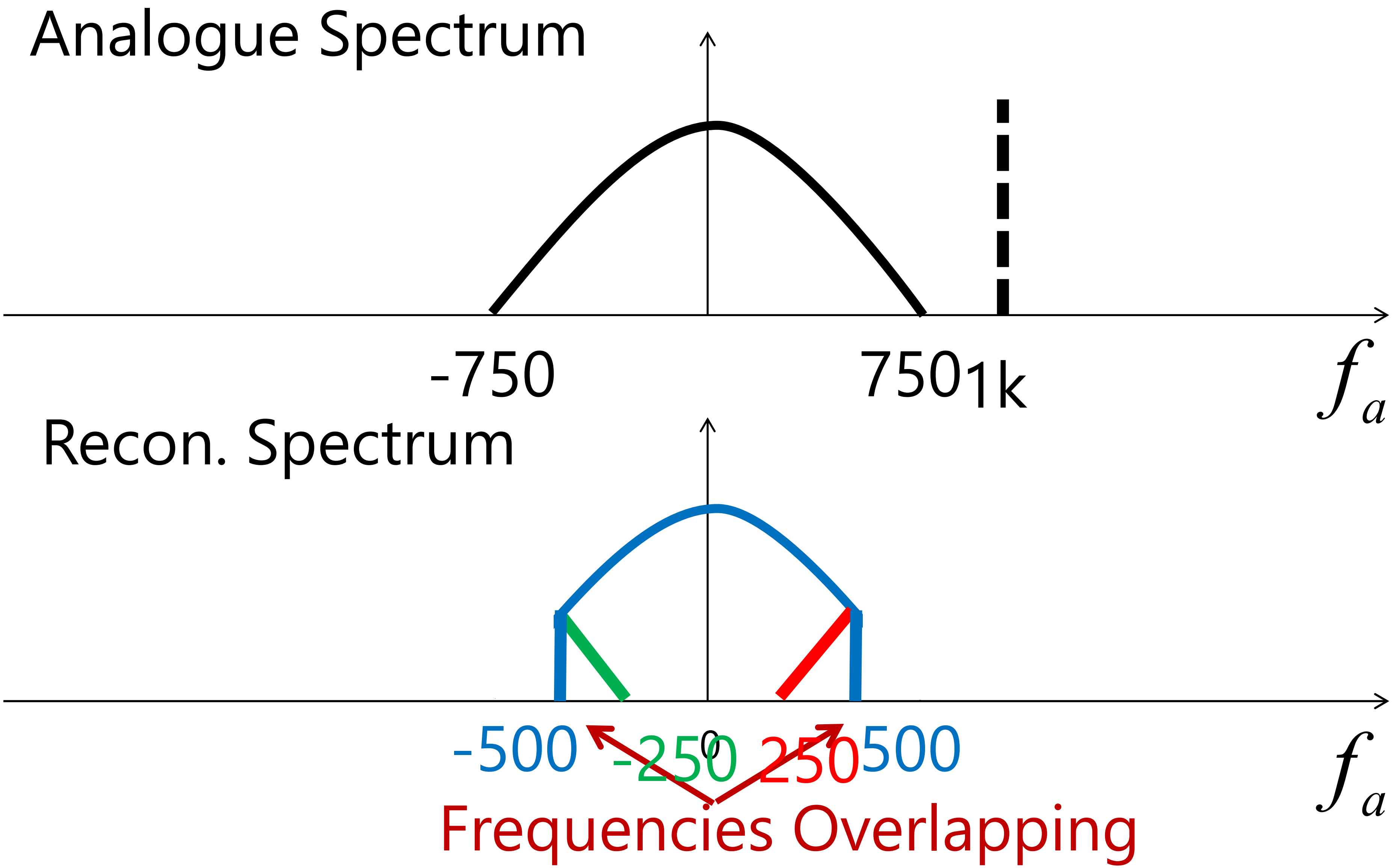


Analogue Spectrum
-750
750
1k
f_a
Digital Spectrum
Disc. Signal
-3.5pi
-1.5pi
-.5pi
0
.5pi
1.5pi
2pi
3.5pi
ω
Frequencies Overlapping

Reconstruction with Alias



Reconstruction with Alias



Example 3.24: Determine Ambiguity Test

- Suppose we are sampling a sinusoidal signal x(t) with has frequency components of signals

$$x(t) = A\cos(\Omega t) \qquad -500\pi \le \Omega \le 500\pi$$

$$x_{\min}(t) = A_{\min}\cos(2\pi(-250)t)$$

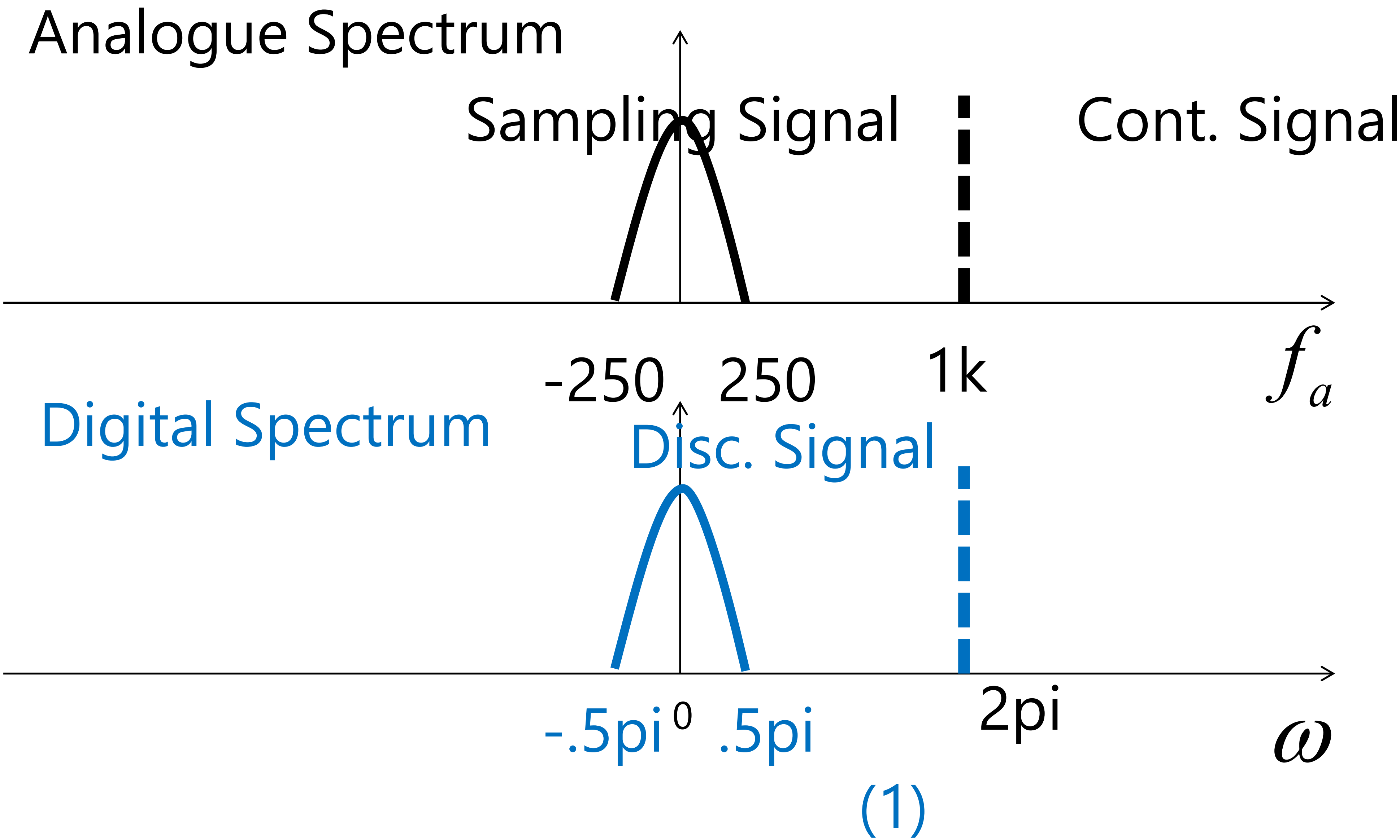
$$x_{\max}(t) = A_{\max}\cos(2\pi(250)t)$$

which means fa is the sampling signal fs.

$$(1)\quad x(n)=\cos(2.5\pi n)$$

- From
$$\omega=\Omega T=\Omega\frac{1}{f_s}$$
$$\omega_{\min}=2\pi f_{a,\min}\frac{1}{f_s}=-500\pi\frac{1}{1000}=-0.5\pi$$
$$\omega_{\max}=2\pi f_{a,\max}\frac{1}{f_s}=500\pi\frac{1}{1000}=0.5\pi$$

- So we have
$$x_{\min}(n)=A_{\min}\cos(-0.5\pi n)$$
$$x_{\max}(n)=A_{\max}\cos(0.5\pi n)$$


Analogue Spectrum
Sampling Signal
Cont. Signal
-250
250
1k
f_a
Digital Spectrum
Disc. Signal
-.5pi
0
.5pi
2pi
ω
(1)

$$x_{\min}(n) = \cos(-1.5\pi n), x_{\max}(n) = \cos(0.5\pi n)$$

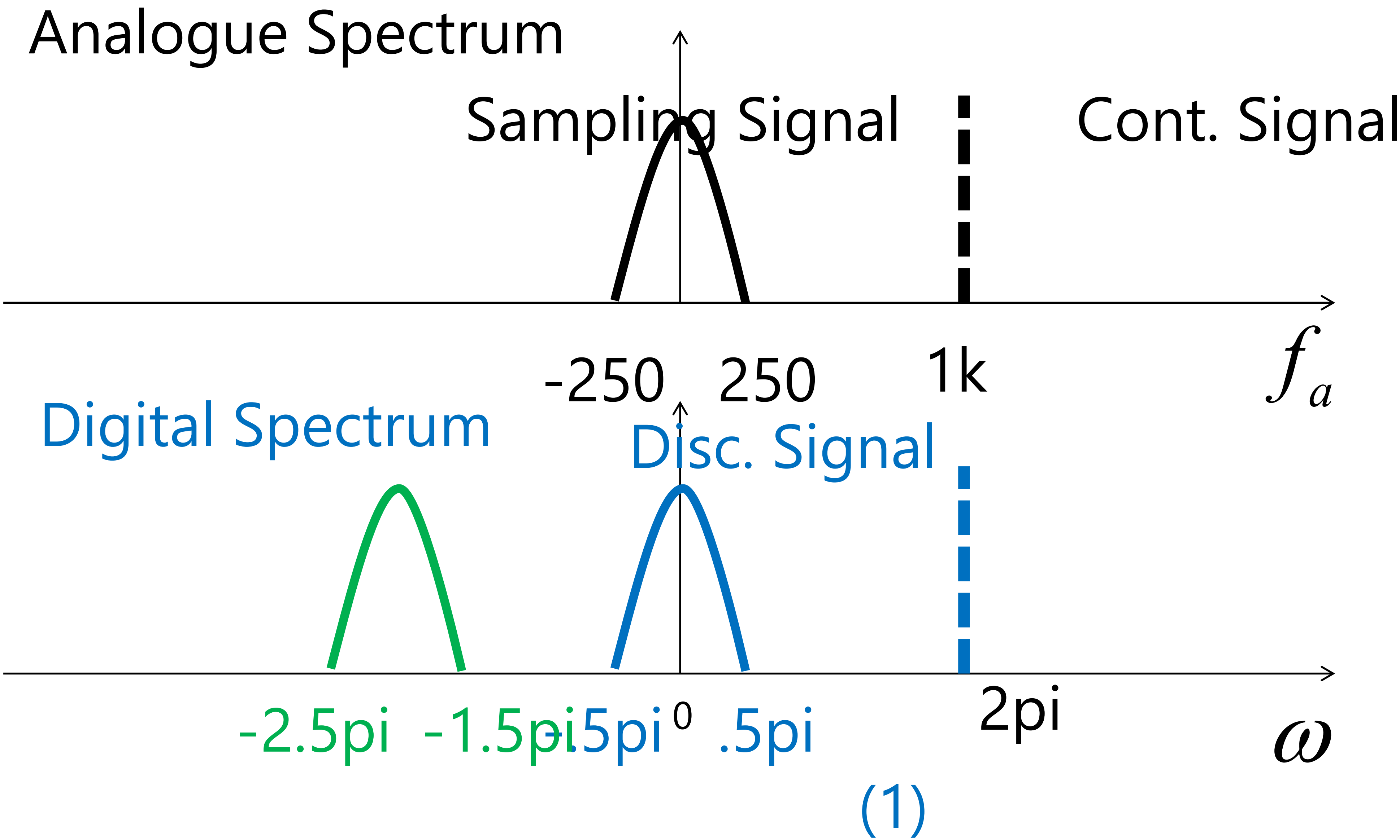
- We can show that

$$\cos(-2.5\pi n) = \cos((-2-0.5)\pi n)$$
$$= \cos(-2\pi n - 0.5\pi n)$$
$$= \cos(-2\pi n)\cos(-0.5\pi n)$$
$$- \sin(-2\pi n)\sin(-0.5\pi n)$$

$$\cos(-1.5\pi n) = \cos((-2+0.5)\pi n)$$
$$= \cos(-2\pi n + 0.5\pi n)$$
$$= \cos(-2\pi n)\cos(0.5\pi n)$$
$$- \sin(-2\pi n)\sin(0.5\pi n)$$

$$\therefore \cos(-2.5\pi n) = \cos(-0.5\pi n)$$

$$\therefore \cos(-1.5\pi n) = \cos(0.5\pi n)$$



Analogue Spectrum
Sampling Signal
Cont. Signal
-250
250
1k
f_a
Digital Spectrum
Disc. Signal
-2.5pi
-1.5pi
-.5pi
0
.5pi
2pi
ω
(1)

$$x_{\min}(n) = \cos(-1.5\pi n), x_{\max}(n) = \cos(0.5\pi n)$$

- We can show that

$$\cos(1.5\pi n) = \cos((2-0.5)\pi n)$$

$$= \cos(2\pi n - 0.5\pi n)$$

$$= \cos(2\pi n)\cos(-0.5\pi n) - \sin(2\pi n)\sin(-0.5\pi n)$$

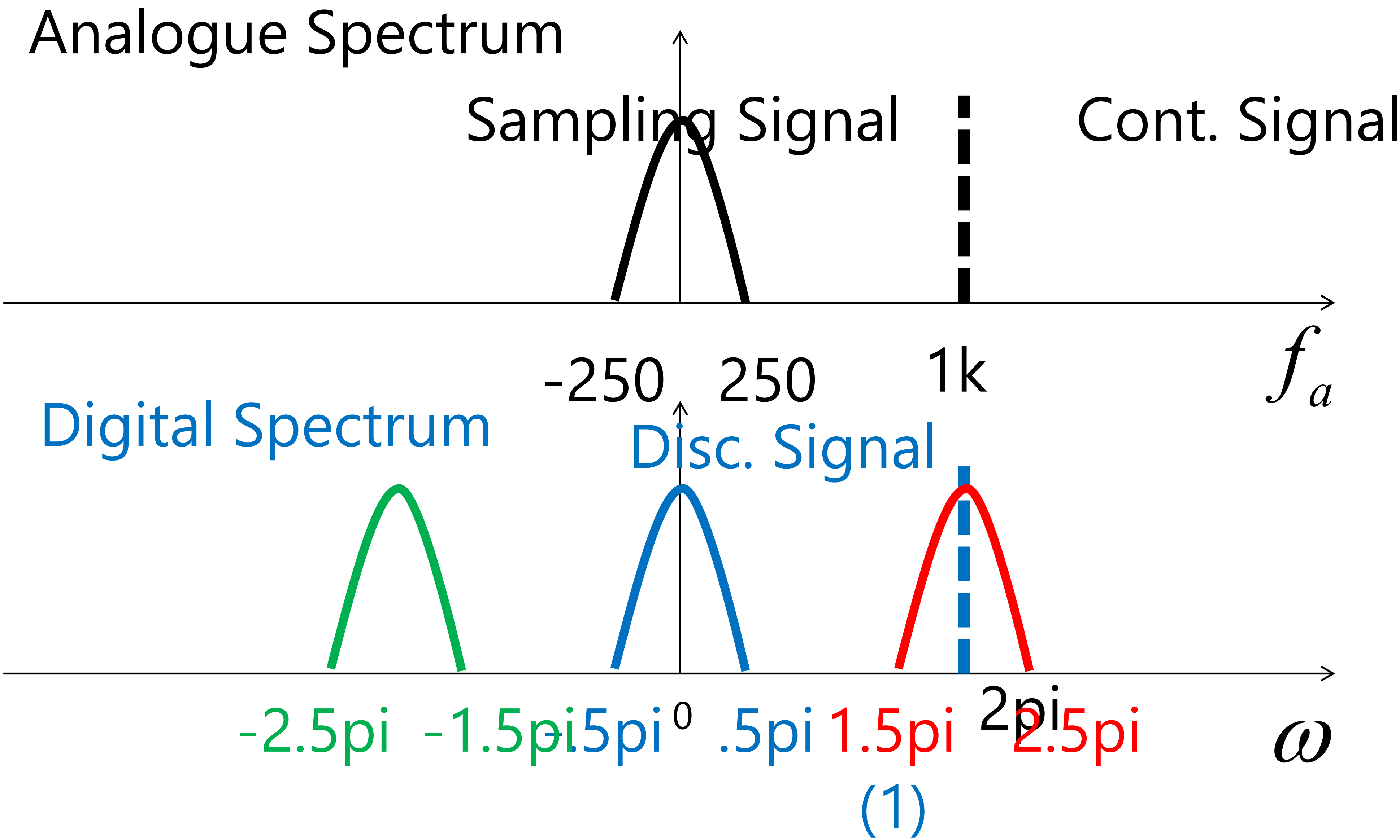
$$\therefore \cos(1.5\pi n) = \cos(-0.5\pi n)$$

$$\cos(2.5\pi n) = \cos((2+0.5)\pi n)$$

$$= \cos(2\pi n + 0.5\pi n)$$

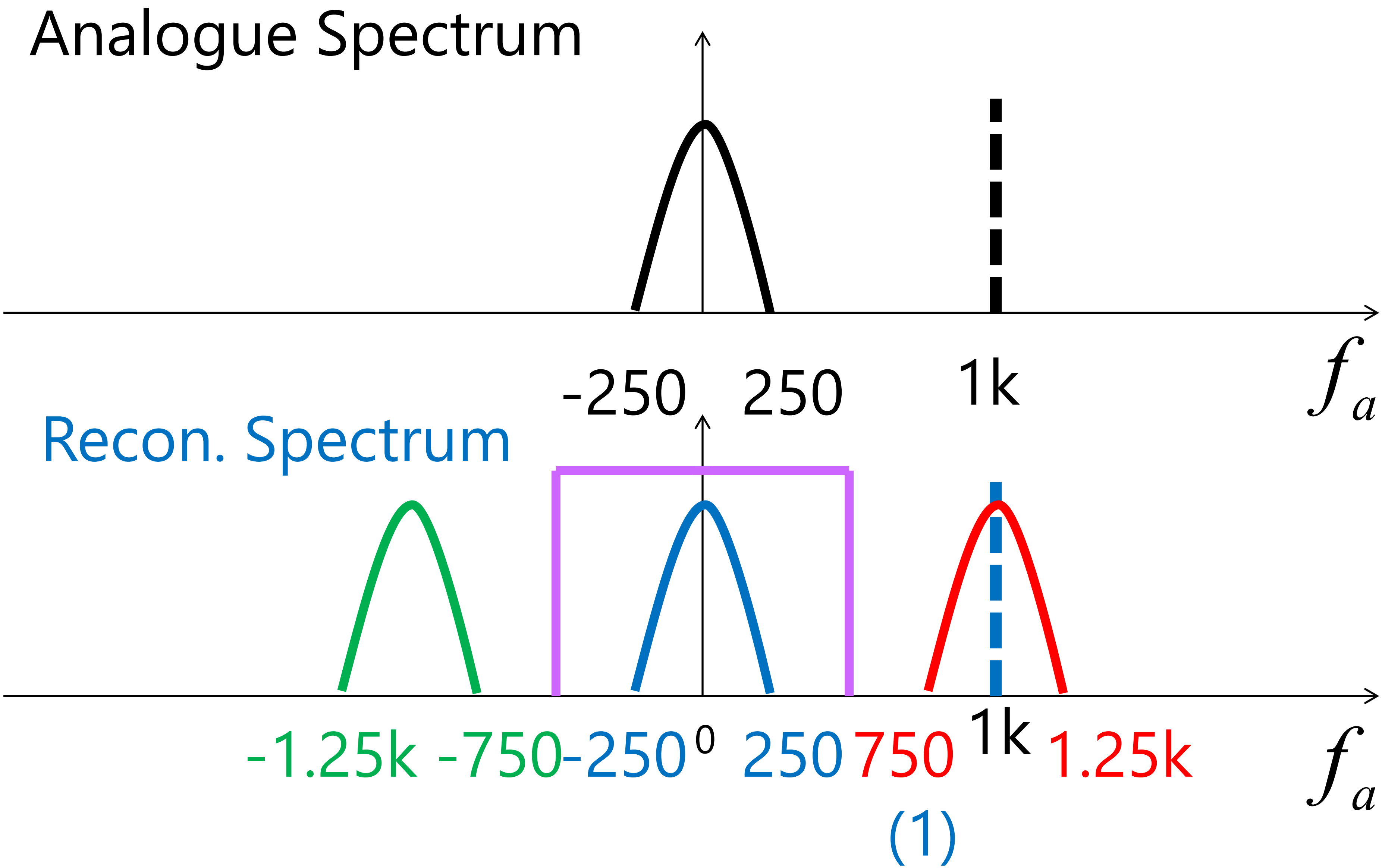
$$= \cos(2\pi n)\cos(0.5\pi n) - \sin(2\pi n)\sin(0.5\pi n)$$

$$\therefore \cos(2.5\pi n) = \cos(0.5\pi n)$$

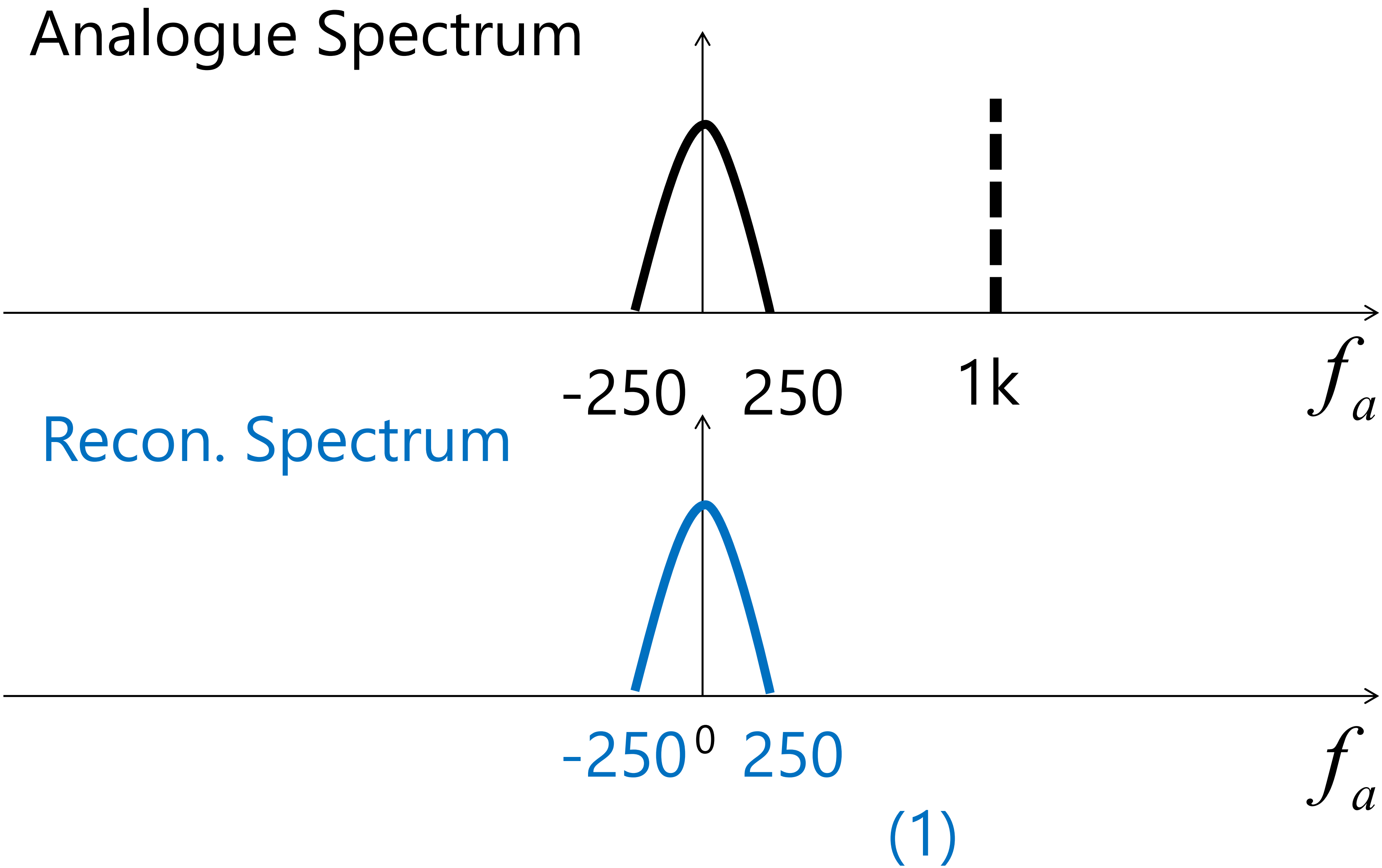


Analogue Spectrum
Sampling Signal
Cont. Signal
-250
250
1k
f_a
Digital Spectrum
Disc. Signal
-2.5pi
-1.5pi
-.5pi
0
.5pi
1.5pi
2pi
2.5pi
(1)
ω

Perfect Reconstruction



Perfect Reconstruction



To Conclude

- The sampling frequency must be greater than twice the maximum frequency of the analogue signal to be sampled

$$2f_{a,\max} < f_s$$

Replicas of spectrum

- In fact, there are *endless* copies of spectra.
- Each locates at multiple of 2pi radians/sample.